COLLEGE SPORTS & RECOVERY TRACKER USING ZOHO CREATOR

A PROJECT REPORT

Submitted by

GOWTHAMRAJ M

2116240701155

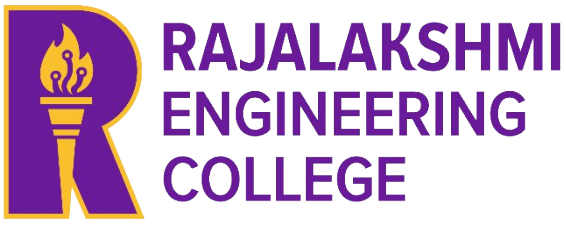
in partial fulfilment of the End Semester Examination for the course

CB23F35 –

LOW-CODE APPLICATION DEVELOPMENT USING ZOHO CREATOR

in

BACHELOR OF ENGINEERING IN COMPUTER SCIENCE ENGINEERING



RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Rajalakshmi Nagar, Thandalam, Tamil Nadu – 602 105

NOVEMBER 2026

RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution, Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this project report titled "College Sports & Recovery Tracker using Zoho Creator" is the bonafide work of GOWTHAMRAJ M (240701155), who carried out the individual project work on the Zoho Creator low-code platform for the course CB23F35 – Low-Code Application Development using Zoho Creator, during the academic year 2026–2027, Semester V, under my supervision.

SIGNATURE
Dr. Manoranjini J
HEAD OF THE DEPARTMENT
Department of Computer Science Engineering
Rajalakshmi Engineering College
Thandalam – 602 105

SIGNATURE
Ms.Aswana Lal
COURSE FACULTY-Assistant Proffessor
Department of Computer Science Engineering
Rajalakshmi Engineering College

Submitted for the End Semester Examination (Individual Project) held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

DECLARATION

I, GOWTHAMRAJ M (2116240701155), hereby declare that the project report titled "College Sports & Recovery Tracker using Zoho Creator", submitted for the End Semester Examination of the course CB23F35 – Low-Code Application Development using Zoho Creator, is a record of original work carried out by me individually. The Zoho Creator application described in this report — including its data model, forms, reports, pages, Deluge scripts, workflows and integrations — was designed and developed by me.

I further declare that this work has not been submitted, in part or in full, for the award of any other degree or course credit. Every template, sample application, AI-assisted feature (such as Zia or the AI application builder), code snippet or external resource used in this work has been duly acknowledged in the report.

Place: Chennai
Date:

SIGNATURE OF THE STUDENT
GOWTHAMRAJ M
2116240701155

ACKNOWLEDGEMENT

I express my sincere gratitude to our Chairman Thiru.S.Meganathan, Chairperson Dr.(Mrs.)Thangam Meganathan and Vice-Chairman Mr.AbhayShankar Meganathan for providing the infrastructure and facilities that made this work possible.

I am grateful to our Principal, Dr.S.N.Murugesan, Rajalakshmi Engineering College, for the constant encouragement extended to students to take up industry-relevant learning.

I thank Dr. Manoranjini J, Head of the Department, Department of Computer Science Engineering, for the guidance and support provided throughout the course.
I owe my sincere thanks to my course faculty, Ms. Aswana Lal , Assistant Professor , for the valuable guidance, reviews and feedback that shaped this application and report.
I also acknowledge Zoho Corporation for providing access to the Zoho Creator platform and learning resources, and I thank my family and friends for their unwavering support.

GOWTHAMRAJ M

ABSTRACT

College sports teams train and compete throughout the year, and injuries are a routine part of that. Yet in most colleges the handling of an injured player — recording the injury, prescribing recovery exercises, monitoring pain and progress, and deciding when the player is fit to return — happens through verbal conversations, notebooks and WhatsApp messages between players, coaches, physical directors and physiotherapists. There is no single record of who is injured, how serious the injury is, whether the recovery plan is on schedule, or whether a player was formally cleared before returning to play, which puts player safety at risk.

This project presents the College Sports & Recovery Tracker, a web application built on the Zoho Creator low-code platform. Five related forms — Player Registration, Injury Report, Recovery Plan, Recovery Progress and Fitness Clearance — capture the complete injury-to-clearance lifecycle using twelve Zoho Creator field types, auto-numbered record IDs, lookup relationships, an image field for injury photos and a Recovery Activities subform that lists the exercises in each plan. Three form workflows provide automatic alerts: an e-mail on every new injury report, a Deluge script that e-mails the coordinator when a recovery plan is past its target end date without being completed, and a Deluge script that raises a high-pain alert whenever a progress entry records a pain level of eight or above. A Sports Recovery Dashboard page built with the UI Builder shows live counts of registered players, recovery plans, injuries and fitness-cleared players, quick links to all five reports, and a pie chart of injury reports.

The application was tested in the Zoho Creator Development environment against functional test cases covering record creation, lookups, subform entry, alert conditions and dashboard updates. It replaces informal, untraceable communication with a structured record in which every injury is linked to its recovery plan, progress history and clearance decision, and in which overdue recoveries and high pain are flagged automatically.

Keywords: Low-Code Development; Zoho Creator; Deluge Scripting; Workflow Automation; Sports Injury Management; Rehabilitation Tracking

PROJECT AT A GLANCE

Attribute	Details
Project Title	College Sports & Recovery Tracker using Zoho Creator
Application Domain	Education / Sports Medicine – College Sports Injury and Rehabilitation Management
Student Name & Register No.	GOWTHAMRAJ M – 2116240701155
Zoho Creator Edition	Free Trial (zoho.in data centre)
Application Name (Link Name)	College Sports & Recovery Tracker (college-sports-recovery-tracker)
Published / Live URL	https://creator.zoho.in/userhome/kaileshm65552/admindashboard#/environments
Demo Credentials (per role)	Administrator: application owner account (shared at demonstration) | No additional roles configured
Number of Forms / Subforms	5 / 1 (Recovery_Activities in Recovery Plan)
Number of Reports (by type)	List: 5 Kanban: 0 Calendar: 0 Chart: 1 (Pie) Others: 0
Number of Pages / Dashboards	1 (Sports_Recovery_Dashboard)
Number of Workflows	Form: 3 Schedule: 0 Approval: 0 Blueprint: 0 Batch: 0
Deluge Functions / Custom Actions	2 Deluge scripts inside form workflows; no custom functions
Integrations	Zoho Mail through the Send Mail workflow action and the Deluge sendmail task
Roles & Profiles Defined	Administrator (application owner) only
Environments Used	Development

Attribute	Details
Date of Production Deployment	Not yet deployed to Production (application runs in the Development environment)

TABLE OF CONTENTS

ABSTRACT iv
PROJECT AT A GLANCE v
LIST OF TABLES ix
LIST OF FIGURES x
LIST OF ABBREVIATIONS xi
INTRODUCTION 1
1.1 Background 1
1.2 Problem Statement 1
1.3 Objectives 2
1.4 Scope of the Project 2
1.5 Justification for a Low-Code Approach 3
1.6 Organisation of the Report 3
LITERATURE SURVEY AND SYSTEM STUDY 5
2.1 Study of the Existing System 5
2.2 Limitations of the Existing System 6
2.3 Low-Code versus Traditional Development 6
2.4 Literature Survey 7
2.5 Emerging Trends in Low-Code Development 8
2.6 Proposed System 9
REQUIREMENT ANALYSIS 10
3.1 Stakeholders and User Roles 10
3.2 Functional Requirements 10
3.3 Non-Functional Requirements 12
3.4 Hardware and Software Requirements 12
3.5 Use Case Model 12
SYSTEM DESIGN 15
4.1 System Architecture 15
4.2 Data Model 16
4.3 Form Design 17
4.4 Process and Workflow Design 20
4.5 User Interface and Navigation Design 21
4.6 Access Control Design 22
IMPLEMENTATION 23
5.1 Application Setup 23
5.2 Forms and Data Capture 23

5.3 Reports 27
5.4 Pages and Dashboards 28
5.5 Deluge Scripting 29
5.6 Workflows and Automation 31
5.7 Integrations 34
5.8 User Management and Role-Based Access 35
5.9 AI and Analytics Features 35
5.10 Pro-Code Extension 36
5.11 Mobile and Portal Access 36
TESTING AND DEBUGGING 37
6.1 Testing Strategy 37
6.2 Test Cases 37
6.3 Debugging Deluge Scripts 39
6.4 Defect Log 39
6.5 User Acceptance Testing 40
DEPLOYMENT, SECURITY AND GOVERNANCE 41
7.1 Environments and Deployment Strategy 41
7.2 Performance Optimisation 42
7.3 Security Features 42
7.4 Governance and Administration 43
RESULTS AND DISCUSSION 44
8.1 Application Output 44
8.2 Achievement of Objectives 45
8.3 Comparison with the Existing System 46
8.4 Limitations of the Low-Code Approach 46
8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum 47
CONCLUSION AND FUTURE ENHANCEMENTS 48
9.1 Conclusion 48
9.2 Future Enhancements 48
9.3 Learning Outcomes and Career Relevance 48
REFERENCES 50
APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST 51
APPENDIX B – COURSE OUTCOME MAPPING 53
APPENDIX C – COMPLETE DELUGE SOURCE CODE 54
APPENDIX D – USER MANUAL 55
D.1 Accessing the Application 55
D.2 Instructions for Sports Coordinator / Administrator 55

D.3 Instructions for Physiotherapist.....55
APPENDIX E – EVALUATION RUBRIC.....57
APPENDIX F – VIVA-VOCE RECORD SHEET.....59

LIST OF TABLES

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application 6
Table 2.2: Summary of Literature Survey 7
Table 3.1: Stakeholders and User Roles 10
Table 3.2: Functional Requirements 10
Table 3.3: Non-Functional Requirements 12
Table 3.4: Hardware and Software Requirements 12
Table 3.5: Use Case Description – Report Injury 13
Table 3.6: Use Case Description – Create Recovery Plan 13
Table 3.7: Use Case Description – Log Recovery Progress 14
Table 4.1: Entities and Relationships 16
Table 4.2: Field Specification – Player_Registration Form 17
Table 4.3: Field Specification – Injury Report Form 17
Table 4.4: Field Specification – Recovery Plan Form 18
Table 4.5: Field Specification – Recovery_Progress Form 19
Table 4.6: Field Specification – Fitness_Clearance Form 19
Table 4.7: Role–Permission Matrix 22
Table 5.1: Reports Implemented 28
Table 5.2: Deluge Scripts Implemented 29
Table 5.3: Workflows Implemented 32
Table 5.4: Integrations Implemented 34
Table 6.1: Test Cases 37
Table 6.2: Defect Log 39
Table 6.3: User Acceptance Feedback 40
Table 7.1: Deployment Checklist 41
Table 7.2: Performance Optimisation Results 42
Table 7.3: Security Controls Implemented 42
Table 8.1: Mapping of Objectives to Outcomes 45
Table 8.2: Existing System versus Proposed Application 46
Table A.1: Feature Coverage Checklist 51
Table B.1: Mapping of Course Outcomes to Report Chapters 53
Table E.1: Evaluation Rubric for the Individual Project 57
Table E.2: Performance Level Descriptors 58
Table F.1: Viva-Voce Questions and Responses 59
Table F.2: Marks Summary 59

LIST OF FIGURES

Figure 2.1: As-Is Process Flow of the Existing Injury Handling Practice.....5
Figure 3.1: Use Case Diagram of the College Sports & Recovery Tracker.....13
Figure 4.1: System Architecture of the College Sports & Recovery Tracker.....15
Figure 4.2: Entity–Relationship Diagram of the Application.....16
Figure 4.3: Injury-to-Clearance Process Flow with Alert Workflows.....20
Figure 4.4: Decision Logic of the Deluge Alert Scripts DS-01 and DS-02.....21
Figure 4.5: Navigation Structure and Wireframe of the Sports_Recovery_Dashboard Page.....21
Figure 5.1: Application Navigation Sections with the Player_Registration Form.....23
Figure 5.2: Player_Registration Form in the Form Builder.....25
Figure 5.3: Injury Report Form in the Form Builder.....25
Figure 5.4: Recovery Plan Form with the Recovery_Activities Subform.....26
Figure 5.5: Recovery_Progress Form in the Form Builder.....26
Figure 5.6: Fitness_Clearance Form in the Form Builder.....27
Figure 5.7: Player_Registration Form Layout in the Design Builder (Web Preview)27
Figure 5.8: Report Navigation Buttons in the Sports Recovery Center Panel.....28
Figure 5.9: Sports_Recovery_Dashboard Page in the Page Builder.....29
Figure 5.10: DS-01 in the Recovery_Overdue_Alert Workflow.....31
Figure 5.11: DS-02 in the Recovery_Progress_Update Workflow.....31
Figure 5.12: Injury_Reported_Alert Workflow (Send Mail Action).....32
Figure 5.13: Form Workflow List – Injury Report.....33
Figure 5.14: Form Workflow List – Recovery Plan.....33
Figure 5.15: Form Workflow List – Recovery_Progress.....34
Figure 5.16: Zia Build Agent (AI Developer) Panel Available in the Application Builder.....36
Figure 6.1: Alert E-mail Received During Testing.....39
Figure 7.1: Application Builder Toolbar Showing the Development Environment (Access Development Live).....41
Figure 8.1: Scenario Step 1 – Injury Recorded in the Injury Report Form.....44
Figure 8.2: Scenario Step 2 – Recovery Plan with Activities in the Subform.....45
Figure 8.3: Scenario Step 5 – Dashboard KPI Panels Reflect Players, Plans, Injuries and Clearances.....45

LIST OF ABBREVIATIONS

Abbreviation	Expansion
API	Application Programming Interface
CRUD	Create, Read, Update, Delete
DS	Deluge Script (script identifier prefix)
ER	Entity–Relationship
FK	Foreign Key
FR	Functional Requirement
KPI	Key Performance Indicator
LCAP	Low-Code Application Platform
NFR	Non-Functional Requirement
PK	Primary Key
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
Zia	Zoho’s AI assistant

CHAPTER 1
INTRODUCTION

1.1 Background

Rajalakshmi Engineering College fields teams in a range of sports — cricket, football, basketball, volleyball, athletics and others — that train regularly and compete in inter-college, zonal and university tournaments. Each team is supported by the Department of Physical Education, with a physical director or sports coordinator responsible for the players and, where available, a physiotherapist who treats injuries. Over a season a single team can see many injuries of different kinds — sprains, strains, fractures, contusions and overuse injuries — affecting different body parts and with different levels of severity.

When a player is injured, the same sequence follows every time. The injury is reported to the coach or physical director; a physiotherapist assesses it and prescribes a recovery plan made up of exercises with a set duration and number of sessions per week; the player follows the plan while progress, pain and completed exercises are monitored; and finally the player is evaluated and either cleared to return to play or advised further rest. Today this sequence is carried out almost entirely through conversations, handwritten notes and messaging apps.

The people involved are the players themselves, the sports coordinator or physical director, who is accountable for the team and for player safety, and the physiotherapist, who designs and supervises the recovery. The information they handle — players, injuries, recovery plans with their activities, progress entries and clearance decisions — is highly relational: every recovery plan belongs to an injury, every injury to a player, and every clearance decision must be traceable to the injury it closes. This makes the process a strong candidate for a structured application.

1.2 Problem Statement

Sports injuries at the college are recorded and followed up informally, so there is no single, reliable record linking a player’s injury to the recovery plan prescribed, the progress made and the decision to clear the player for play. Recoveries that run past their target date and players reporting severe pain go unnoticed until someone happens to ask, and players may return to play without a documented fitness clearance.

- **No central injury record:** injuries are noted in notebooks or chat messages, so the number, type and severity of injuries per sport or per season cannot be known.
- **Unstructured recovery plans:** exercises, durations and sessions per week are communicated orally or on paper and are easily forgotten or changed.
- **Unmonitored progress:** pain level and percentage of recovery are not recorded consistently, so a worsening condition is not detected early.
- **Missed deadlines:** nobody is alerted when a recovery plan passes its target end date without being completed.
- **Undocumented return to play:** fitness clearance decisions are made verbally, with no record of who evaluated the player, the result or the recommendation.

1.3 Objectives

- To design a relational data model and five Zoho Creator forms (Player Registration, Injury Report, Recovery Plan, Recovery Progress and Fitness Clearance) linked by lookups, with auto-numbered IDs and a Recovery Activities subform.
- To automate an e-mail alert to the sports coordinator whenever a new injury is reported.
- To automate detection of overdue recovery plans and high pain levels using Deluge scripts that send alert e-mails.
- To provide a Sports Recovery Dashboard with live KPI counts, quick access to all reports and a chart of injury reports.
- To test the application systematically in the Development environment and document its deployment, security and governance path to production.

1.4 Scope of the Project

In scope. The application covers the complete injury lifecycle for college sports teams: registering players with their department, year, sport, team and contact details; reporting injuries with date, sport, body part, injury type, severity, cause, description, photo and status; creating recovery plans with start and target end dates, the physiotherapist and a list of recovery activities (activity name, duration, sessions per week, instructions and status); logging recovery progress with pain level, progress percentage, exercise completion and physiotherapist remarks; and recording fitness

clearance with evaluation date, result, recommendation, reviewer and approval status. Automation covers injury alerts, overdue-recovery alerts and high-pain alerts by e-mail, and a live dashboard. The application is accessed through the Zoho Creator web interface and is available on the Zoho Creator mobile app without separate development.

User roles. The design defines two roles — Sports Coordinator/Administrator and Physiotherapist. In the current build the application is operated through the Administrator (application owner) account; the Physiotherapist permission profile is designed in Section 4.6 and listed as a future enhancement.

Out of scope. Medical diagnosis, medical-imaging storage beyond a single injury photo, insurance claims, integration with hospital systems, scheduling of physiotherapy appointments and a player-facing portal are outside the scope of this project.

1.5 Justification for a Low-Code Approach

Unit I of the course describes low-code development as building applications chiefly through visual configuration, adding code only where the platform's built-in components are not sufficient [1]. Injury and recovery tracking fits that description well. The problem is dominated by structured data capture — forms, relationships, a list of activities per plan, status fields — and by a handful of rules that trigger notifications, all of which Zoho Creator provides through its form builder, subforms, lookups, page builder, form workflows and the Deluge scripting language [2].

A low-code approach also suits the people who will maintain the system. The physical education department does not have a software team, and staff change over time; forms and alerts that can be adjusted visually are far easier to maintain than custom code. Zoho Creator hosts the application, handles sign-in and provides a mobile app automatically, so a coach or physiotherapist can open a player's recovery plan on a phone at the ground without any server or deployment work. Finally, the speed of building — five forms, three workflows and a dashboard assembled in a matter of days — means the application can be extended season by season as new needs appear.

1.6 Organisation of the Report

Chapter 2 reviews the existing system and related low-code literature. Chapter 3 presents the requirement analysis. Chapter 4 describes the system design. Chapter 5

details the implementation on Zoho Creator. Chapter 6 covers testing and debugging. Chapter 7 discusses deployment, performance optimisation, security and governance. Chapter 8 presents results and discussion, and Chapter 9 concludes the report with future enhancements.

CHAPTER 2
LITERATURE SURVEY AND SYSTEM STUDY

2.1 Study of the Existing System

The existing practice was studied by following how an injury is handled from the moment it occurs until the player returns to play (Figure 2.1). (1) A player is injured during practice or a match and informs the coach verbally. (2) The coach notes the injury in a notebook or sends a WhatsApp message to the physical director. (3) A physiotherapist, if available, assesses the injury and explains recovery exercises orally or on a sheet of paper. (4) The player follows the exercises and reports progress informally — pain levels and completed sessions are rarely written down. (5) When the player feels better, the coach decides on return to play based on conversation, without a formal clearance record. (6) At the end of the season there is no consolidated record of the injuries that occurred, how long recovery took, or which players were cleared and by whom.

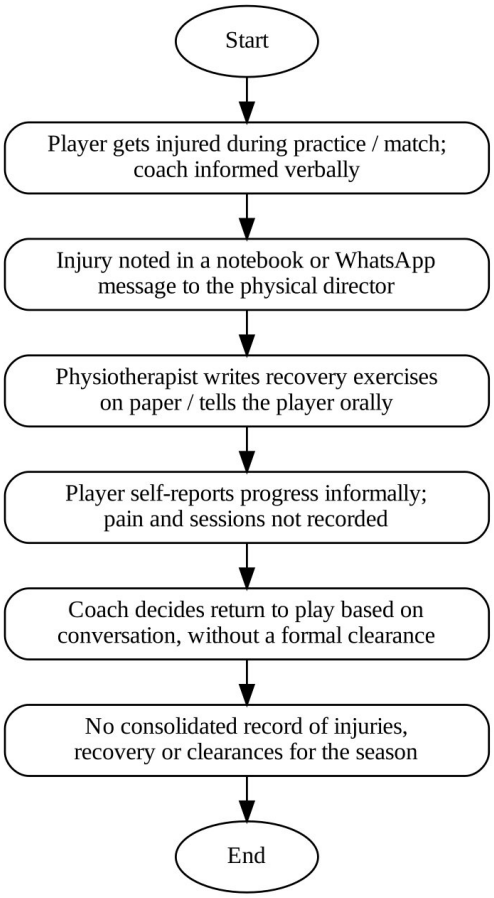


Figure 2.1: As-Is Process Flow of the Existing Injury Handling Practice

2.2 Limitations of the Existing System

- Injury details are scattered across notebooks and chat messages, with no structured fields for body part, injury type, severity or cause.
- Recovery plans are not recorded in a structured form, so the prescribed activities, durations and session frequencies are easily lost or misremembered.
- Progress is not measured consistently, so increasing pain or stalled recovery is not noticed early.
- No one is alerted when a recovery runs past its target date or when a player reports severe pain.
- Return-to-play decisions are undocumented, exposing players to re-injury and the college to accountability gaps.
- Season-level information — injuries per sport, recovery durations, clearances — cannot be produced without manual compilation.

2.3 Low-Code versus Traditional Development

Table 2.1 compares building this application on a traditional web stack (for example, a React front end, a Node.js or Java back end and a relational database) with building it on Zoho Creator, where the alert logic is written in Deluge [3] and everything else is configuration.

Table 2.1: Comparison of Low-Code and Traditional Development for the Proposed Application

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Development time	About 6–8 weeks for one developer: schema, REST API, five CRUD screens, a nested activity list, file upload for photos, e-mail service, dashboard and authentication	About 1–2 weeks: five forms with a subform, auto-generated reports, three form workflows, one dashboard page
Skills required	Front-end framework, back-end language, SQL, REST API design, file storage, authentication, DevOps	Data modelling, Zoho Creator builders, basic Deluge scripting
Hosting & infrastructure	Server or cloud service, database, image storage, domain and SSL certificate to provision and pay for	Hosted by Zoho, including image storage; no infrastructure to manage

Criterion	Traditional (Pro-Code) Development	Low-Code on Zoho Creator
Mobile application	Separate responsive design or mobile app needed for use at the ground	Same application available on the Zoho Creator mobile app automatically
Security & access control	Authentication and authorisation must be coded; medical-type data needs careful handling	Zoho account sign-in and configurable permission profiles built in
Maintenance & change	Every new field or alert needs a developer and redeployment	Fields, drop-down values and alert rules can be changed visually by trained staff
Customisation limits	Unlimited — any UI or logic can be built	Bounded by platform components, page-builder styling and Deluge limits
Cost	Hosting, storage and domain charges plus developer time	Free trial / education edition sufficient for the prototype; per-user plan for production

2.4 Literature Survey

Research literature on low-code platforms and the official Zoho Creator documentation were reviewed. Table 2.2 summarises the sources and their relevance to this project.

Table 2.2: Summary of Literature Survey

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
1	A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, 2020 [4]	Supporting the understanding and comparison of low-code development platforms	Proposes a taxonomy of low-code platform features — graphical UI building, data modelling, business logic, interoperability, security and scalability — and compares commercial platforms against it.	Used as the checklist for mapping each requirement to a Zoho Creator feature (forms, subform, workflows, Deluge, pages).
2	A. C. Bock and U. Frank, 2021 [5]	Low-Code Platform (Business & Information Systems Engineering)	Characterises low-code platforms by their data-modelling, GUI-design, process and integration components, aimed at lowering the barrier to building business applications.	Shaped the layered architecture of the application in Chapter 4.

S.No.	Author(s) / Source & Year	Title / Product	Key Findings	Relevance to this Project
3	D. Di Ruscio et al., 2022 [6]	Low-code development and model-driven engineering: Two sides of the same coin?	Relates low-code platforms to model-driven engineering and discusses challenges such as vendor lock-in, scalability and extensibility.	Informed the discussion of limitations and vendor lock-in in Section 8.4.
4	M. Hirzel, 2023 [7]	Low-code programming models (Communications of the ACM)	Surveys low-code programming models and how visual configuration is combined with small amounts of conventional code.	Framed the no-code / low-code / pro-code split in Section 8.5.
5	Zoho Corporation [2]	Zoho Creator Help Documentation	Documents field types including Auto Number, Image, Decision Box and Subform; lookups; reports; the page builder; form workflows; and the Deluge sendmail task and zoho.currentdate.	Primary implementation reference for every component in Chapter 5.

2.5 Emerging Trends in Low-Code Development

Three trends influenced the design. First, **citizen development** — domain staff building their own tools — is a major driver of the growth reported for enterprise low-code platforms by analysts such as Gartner [8]. A physical director or physiotherapist is exactly such a domain expert, so the application keeps nearly everything in visual configuration and uses Deluge only for the two conditional alerts.

Second, **AI-assisted application building** is now part of the platform. The Zoho Creator builder used for this project includes the Zia “Build Agent”, an AI developer panel that can generate forms and logic from natural-language prompts [9]; it is visible beside every builder screen in Chapter 5 and is discussed in Section 5.9.

Third, **hyper-automation** — automating each repeatable step around a process — led to the decision to make the application proactive: rather than waiting for staff to check records, it e-mails the coordinator on new injuries, overdue recoveries and high pain.

The **composable** design of the platform also allowed the dashboard to reuse existing reports through buttons and a chart instead of a separate analytics module.

2.6 Proposed System

The proposed College Sports & Recovery Tracker replaces notebooks and messages with one Zoho Creator application. Each limitation in Section 2.2 is addressed: injuries are captured in a structured Injury Report with drop-downs for sport, body part, injury type, severity and cause, plus a photo; recovery plans are stored with their activities in a subform, so every exercise, duration and frequency is recorded; Recovery Progress captures pain level, progress percentage and exercise completion over time; Deluge alerts flag overdue plans and high pain automatically, and a Send Mail action announces every new injury; Fitness Clearance documents the evaluation, result, reviewer and approval for every return to play; and the Sports Recovery Dashboard gives live counts and a chart for season-level visibility.

CHAPTER 3
REQUIREMENT ANALYSIS

3.1 Stakeholders and User Roles

The stakeholders of the application and their roles are listed in Table 3.1.

Table 3.1: Stakeholders and User Roles

Role	Description	Key Responsibilities in the Application
Sports Coordinator / Administrator	Physical director or sports coordinator who owns the application	Register players; report injuries; record fitness clearance; view all reports and the dashboard; receive alert e-mails
Physiotherapist	Treats injured players and supervises recovery	Create recovery plans and activities; log progress; recommend fitness clearance (designed role — see Section 4.6)
Player (stakeholder)	Student athlete registered with a team	Subject of the records; follows the recovery plan (no direct login in the current build)
Zoho Creator platform (system)	Hosting, authentication and mail service	Stores data and photos, runs workflows and Deluge scripts, delivers alert e-mails

3.2 Functional Requirements

Table 3.2: Functional Requirements

FR ID	Requirement	Role(s)	Priority
FR-01	The system shall register a player with an auto-generated Player ID, name, register number, department, year, sport, team name, phone number and e-mail.	Admin	High
FR-02	The system shall record an injury against a registered player (lookup) with injury date, sport, body part, injury type, severity, cause, description, photo and injury status.	Admin	High
FR-03	The system shall send an e-mail to the sports coordinator whenever a new injury report is submitted.	System	High
FR-04	The system shall create a recovery plan linked to an injury, with start date, target end date, physiotherapist name, plan status and one or more recovery activities.	Admin, Physio	High
FR-05	The system shall e-mail an overdue alert when a recovery plan is created or edited with a target end date earlier than today and a status other than Completed.	System	High

FR ID	Requirement	Role(s)	Priority
FR-06	The system shall log recovery progress against a plan with progress date, pain level, progress percentage, exercise completion, remarks and current status.	Admin, Physio	High
FR-07	The system shall e-mail a high-pain alert when a progress entry records a pain level of 8 or more.	System	High
FR-08	The system shall record a fitness clearance linked to an injury with evaluation date, fitness result, recommendation, reviewer, approval status and remarks.	Admin, Physio	High
FR-09	The system shall display a dashboard with counts of players, recovery plans, injuries and fitness-cleared players, links to all reports and a chart of injury reports.	Admin	Medium
FR-10	The system shall list all records of each form in a report from which records can be viewed, edited and deleted.	Admin	Medium

3.3 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	Forms shall submit and reports and the dashboard shall load within 3 seconds for up to 1,000 records per form.
NFR-02	Security	Only authenticated Zoho users added to the application shall access it; injury and recovery data shall be visible only to authorised roles.
NFR-03	Usability	The application shall be usable on desktop browsers and on the Zoho Creator mobile app at the sports ground.
NFR-04	Availability	The application shall be available whenever the Zoho Creator cloud service is available, with no self-managed server.
NFR-05	Maintainability	Drop-down values, fields and alert thresholds shall be modifiable through the visual builders and Deluge editor without redeployment.
NFR-06	Traceability	Every recovery plan, progress entry and clearance shall be linked to its injury and player through lookups and unique auto-numbered IDs.

3.4 Hardware and Software Requirements

Table 3.4: Hardware and Software Requirements

Category	Specification
Platform	Zoho Creator (Free Trial edition, zoho.in data centre)
Scripting language	Deluge (Data Enriched Language for the Universal Grid Environment)
Browser	Google Chrome (latest version); Microsoft Edge / Mozilla Firefox also supported
API testing tool	Not required — no external API calls in the current build
Other Zoho services	Zoho Mail (Send Mail action and Deluge sendmail task)
Mobile	Zoho Creator mobile app (Android / iOS) — optional access channel
Client hardware	Intel Core i3/i5 or Apple M-series, 8 GB RAM, internet connection

3.5 Use Case Model

Figure 3.1 shows the use cases for both roles. Three key use cases are described in Tables 3.5 to 3.7.

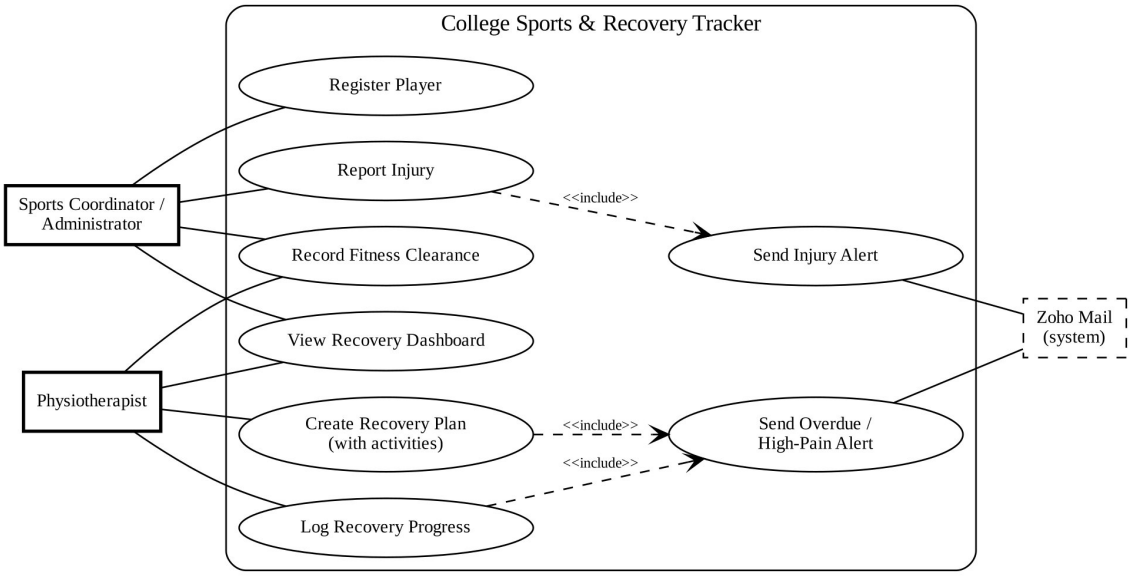


Figure 3.1: Use Case Diagram of the College Sports & Recovery Tracker

Table 3.5: Use Case Description – Report Injury

Use Case ID	UC-01
Use Case Name	Report Injury
Actor(s)	Sports Coordinator / Administrator
Pre-condition	The player exists in Player_Registration; the coordinator is signed in.
Main Flow	1. Open the Injury Report form. 2. Select the player through the Player_Registration lookup. 3. Enter Injury Date and choose Sport, Body Part, Injury Type, Severity and Cause. 4. Enter a Description and upload an Injury Photo. 5. Set Injury Status and Submit. 6. The system assigns an Injury ID and sends the Injury_Reported_Alert e-mail.
Alternate Flow	2a. Player not yet registered: register the player first in Player_Registration, then repeat.
Post-condition	The injury is stored with a unique ID, linked to the player, counted on the dashboard and the coordinator has been e-mailed.
Related FR	FR-02, FR-03, FR-09

Table 3.6: Use Case Description – Create Recovery Plan

Use Case ID	UC-02
Use Case Name	Create Recovery Plan
Actor(s)	Physiotherapist (through the administrator account in the current build)
Pre-condition	The injury exists in Injury Report.

Main Flow	1. Open the Recovery Plan form. 2. Select the Injury Report. 3. Enter Start Date, Target End Date and Physiotherapist Name; set Plan Status. 4. Add one row per exercise in Recovery_Activities (Activity Name, Duration in Minutes, Sessions per Week, Instructions, Activity Status). 5. Submit. 6. The Recovery_Overdue_Alert workflow checks the dates and status.
Alternate Flow	6a. Target End Date is earlier than today and status is not Completed: the system e-mails a “Recovery Overdue Alert”.
Post-condition	The plan and its activities are stored and linked to the injury.
Related FR	FR-04, FR-05

Table 3.7: Use Case Description – Log Recovery Progress

Use Case ID	UC-03
Use Case Name	Log Recovery Progress
Actor(s)	Physiotherapist (through the administrator account in the current build)
Pre-condition	A recovery plan exists for the injury.
Main Flow	1. Open Recovery_Progress. 2. Select the Recovery Plan. 3. Enter Progress Date, Pain Level (0–10), Progress Percentage, tick Exercise Completed, add Physiotherapist Remarks and Current Status. 4. Submit.
Alternate Flow	4a. Pain Level is 8 or more: the Recovery_Progress_Update workflow e-mails a “High Pain Level Alert”.
Post-condition	The progress entry is stored against the plan and visible in the Recovery_Progress Report.
Related FR	FR-06, FR-07

CHAPTER 4
SYSTEM DESIGN

4.1 System Architecture

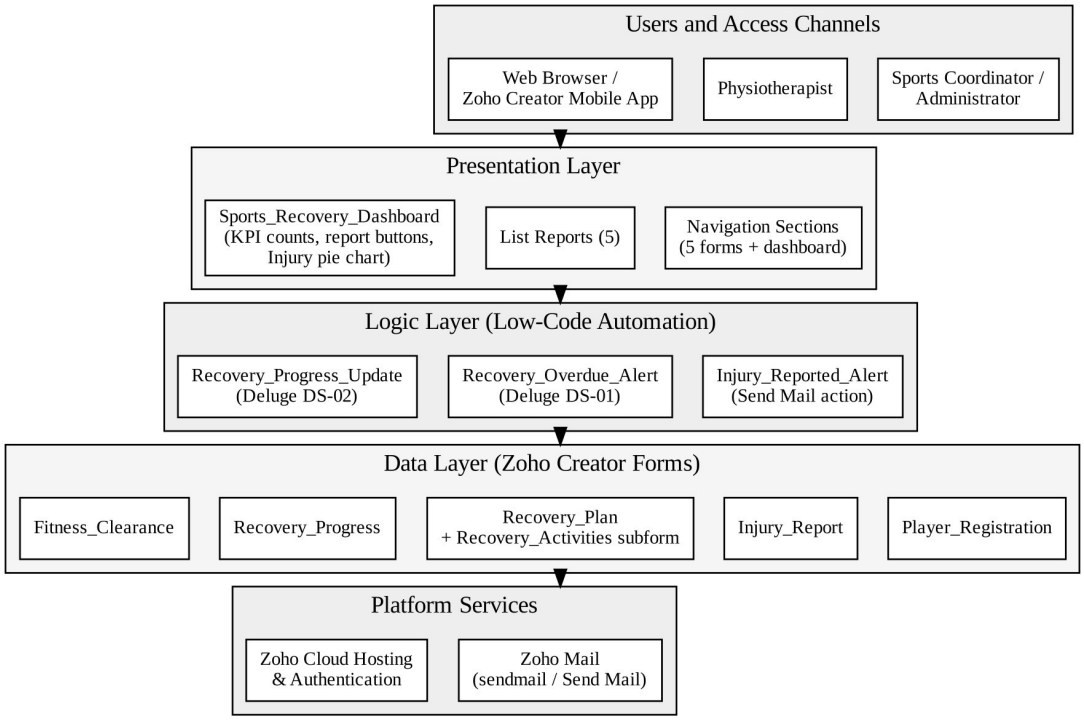


Figure 4.1: System Architecture of the College Sports & Recovery Tracker

Figure 4.1 shows the layered architecture. The **users and access channels** layer consists of the Sports Coordinator/Administrator and the Physiotherapist, who reach the application through a web browser or the Zoho Creator mobile app after signing in with a Zoho account. The **presentation layer** comprises the navigation sections for the five forms, the list report generated for each form, and the Sports_Recovery_Dashboard page.

The **logic layer** holds the low-code automation: the Injury_Reported_Alert workflow with a no-code Send Mail action, and the Recovery_Overdue_Alert and Recovery_Progress_Update workflows, each containing a Deluge script (DS-01 and DS-02). The **data layer** is formed by the five Zoho Creator forms and the Recovery_Activities subform, with relationships realised by lookup fields. The **platform services** layer provides hosting, image storage, authentication and the mail service used by both the Send Mail action and the Deluge sendmail task.

4.2 Data Model

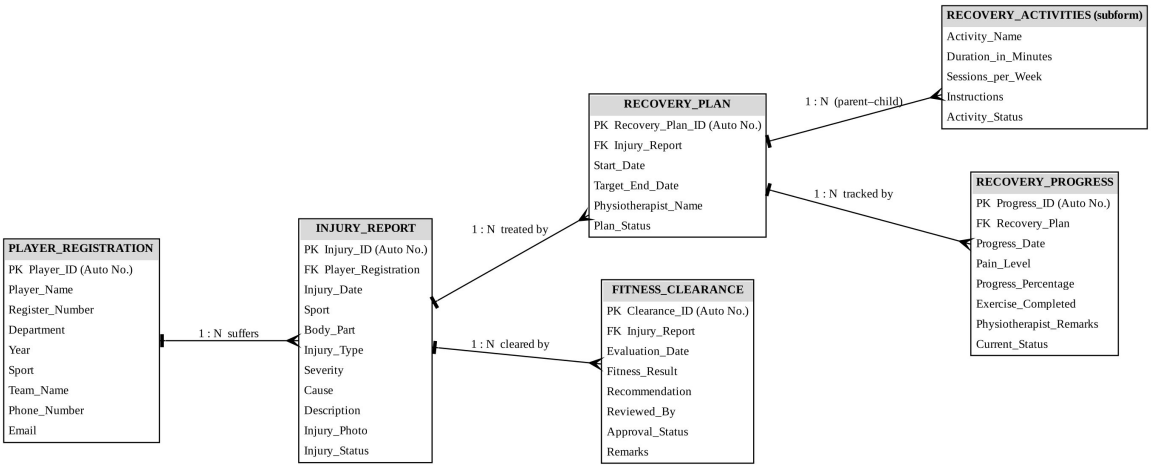


Figure 4.2: Entity–Relationship Diagram of the Application

Figure 4.2 presents the entity–relationship model. Player_Registration is the master entity: a player can suffer many injuries, so Injury_Report holds a lookup to Player_Registration. Each injury can have one or more recovery plans (Recovery_Plan holds a lookup to Injury_Report) and one or more fitness evaluations (Fitness_Clearance holds a lookup to Injury_Report). Each recovery plan contains many activities, implemented as the Recovery_Activities subform — a parent–child relationship in which activity rows exist only within their plan — and is tracked by many progress entries (Recovery_Progress holds a lookup to Recovery_Plan). Every form uses an Auto Number field (Player ID, Injury ID, Recovery Plan ID, Progress ID, Clearance ID) as a human-readable identifier alongside Zoho Creator's internal record ID. Table 4.1 summarises the relationships.

Table 4.1: Entities and Relationships

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Player_Registration	Player_ID, Player_Name, Register_Number, Sport	Injury_Report	One-to-Many	Lookup (Player_Registration) in Injury Report
Injury_Report	Injury_ID, Injury_Date, Severity, Injury_Status	Recovery_Plan	One-to-Many	Lookup (Injury Report) in Recovery Plan
Injury_Report	Injury_ID	Fitness_Clearance	One-to-Many	Lookup (Injury Report) in Fitness_Clearance

Entity (Form)	Key Fields	Related To	Relationship Type	Implemented Using
Recovery_Plan	Recovery_Plan_ID, Target_End_Date, Plan_Status	Recovery_Activities	Parent–Child	Subform in Recovery Plan
Recovery_Plan	Recovery_Plan_ID	Recovery_Progress	One-to-Many	Lookup (Recovery Plan) in Recovery_Progress

4.3 Form Design

The field specification of each form is given in Tables 4.2 to 4.6. Mandatory flags were not enabled on these forms in the current build (see BUG-03 in Section 6.4); the Auto Number fields are generated by the system starting at 1.

Table 4.2: Field Specification – Player_Registration Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Player ID	Player_ID	Auto Number	—	Start index 1	System generated
Player Name	Player_Name	Single Line	No	—	—
Register Number	Register_Number	Single Line	No	—	College register number
Department	Department	Drop Down	No	Department choices	—
Year	Year	Drop Down	No	Year choices	—
Sport	Sport	Drop Down	No	Sport choices	—
Team Name	Team_Name	Single Line	No	—	—
Phone Number	Phone_Number	Phone	No	Country code +91	—
Email	Email	Email	No	E-mail format check	—

Table 4.3: Field Specification – Injury Report Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Injury ID	Injury_ID	Auto Number	—	Start index 1	System generated
Player_Registration	Player_Registration	Lookup	No	Source: Player_Registration	Injured player

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Injury Date	Injury_Date	Date	No	—	Used for season analysis
Sport	Sport	Drop Down	No	Sport choices	—
Body Part	Body_Part	Drop Down	No	e.g., Ankle / Knee / Shoulder	—
Injury Type	Injury_Type	Drop Down	No	e.g., Sprain / Strain / Fracture	—
Severity	Severity	Drop Down	No	e.g., Mild / Moderate / Severe	—
Cause	Cause	Drop Down	No	Cause choices	—
Description	Description	Multi Line	No	—	—
Injury Photo	Injury_Photo	Image	No	Image upload	Visual record of injury
Injury Status	Injury_Status	Drop Down	No	Status choices	—

Table 4.4: Field Specification – Recovery Plan Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Recovery Plan ID	Recovery_Plan_ID	Auto Number	—	Start index 1	System generated
Injury Report	Injury_Report	Lookup	No	Source: Injury Report	Injury being treated
Start Date	Start_Date	Date	No	—	—
Target End Date	Target_End_Date	Date	No	—	Used by DS-01
Physiotherapist Name	Physiotherapist_Name	Single Line	No	—	—
Plan Status	Plan_Status	Drop Down	No	Includes “Completed”	Used by DS-01
Recovery_Activities	Recovery_Activities	Subform	No	One row per activity	Parent–child rows
↳ Activity Name	Activity_Name	Single Line	No	—	Subform field
↳ Duration in Minutes	Duration_in_Minutes	Number	No	Numeric	Subform field

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
↳ Sessions per Week	Sessions_per_Week	Number	No	Numeric	Subform field
↳ Instructions	Instructions	Multi Line	No	—	Subform field
↳ Activity Status	Activity_Status	Drop Down	No	Status choices	Subform field

Table 4.5: Field Specification – Recovery_Progress Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Progress ID	Progress_ID	Auto Number	—	Start index 1	System generated
Recovery Plan	Recovery_Plan	Lookup	No	Source: Recovery Plan	—
Progress Date	Progress_Date	Date	No	—	—
Pain Level	Pain_Level	Number	No	Scale 0–10	DS-02 alerts at 8 or more
Progress Percentage	Progress_Percentage	Number	No	0–100	—
Exercise Completed	Exercise_Completed	Decision Box	No	Checked / unchecked	—
Physiotherapist Remarks	Physiotherapist_Remarks	Multi Line	No	—	—
Current Status	Current_Status	Drop Down	No	Status choices	—

Table 4.6: Field Specification – Fitness_Clearance Form

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Clearance ID	Clearance_ID	Auto Number	—	Start index 1	System generated
Injury Report	Injury_Report	Lookup	No	Source: Injury Report	Injury being closed
Evaluation Date	Evaluation_Date	Date	No	—	—
Fitness Result	Fitness_Result	Drop Down	No	e.g., Fit / Not Fit	—
Recommendation	Recommendation	Multi Line	No	—	—
Reviewed By	Reviewed_By	Single Line	No	—	Evaluator
Approval Status	Approval_Status	Drop Down	No	e.g., Approved / Pending	Feeds “fitness cleared” KPI

Field Label	Link Name	Field Type	Mandatory	Validation / Default	Remarks
Remarks	Remarks	Multi Line	No	—	—

4.4 Process and Workflow Design

Figure 4.3 shows the end-to-end injury lifecycle supported by the application and the points at which each alert workflow runs.

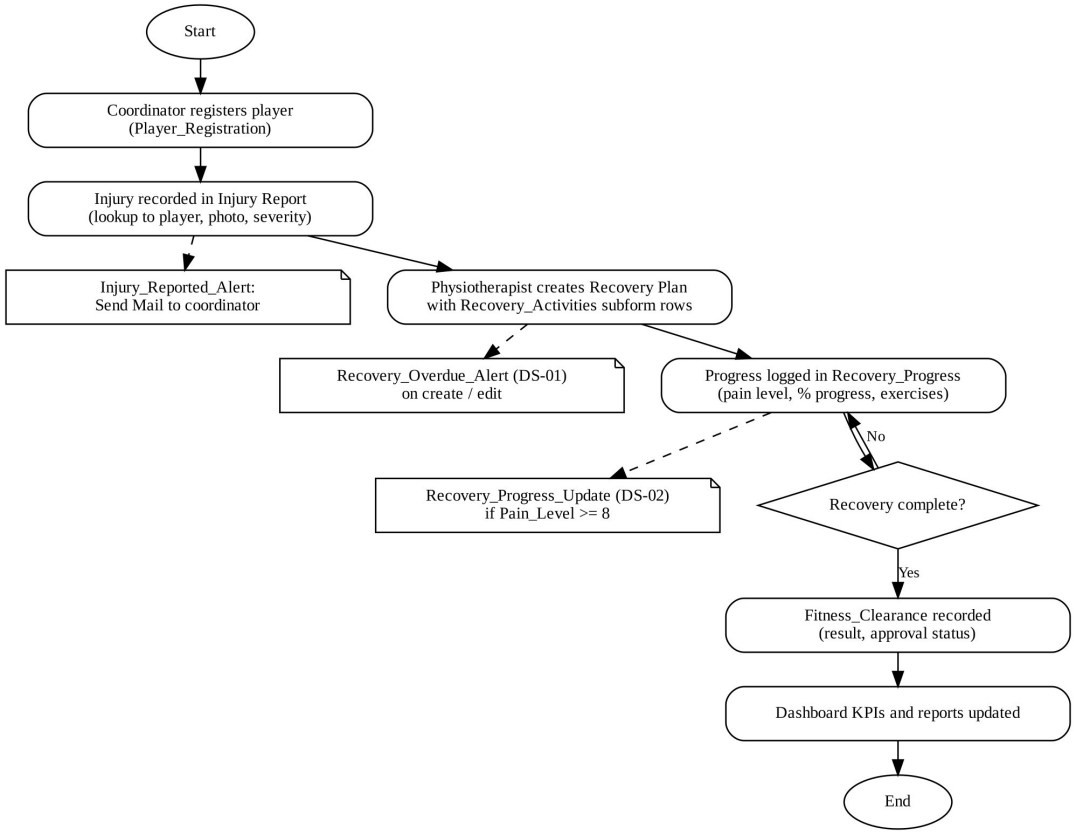


Figure 4.3: Injury-to-Clearance Process Flow with Alert Workflows

Figure 4.4 shows the decision logic of the two Deluge alert scripts. Both run in the "Successful form submission" stage of a "Created or Edited" form workflow, so they are evaluated every time a record is saved, and both send mail only when their condition is true.

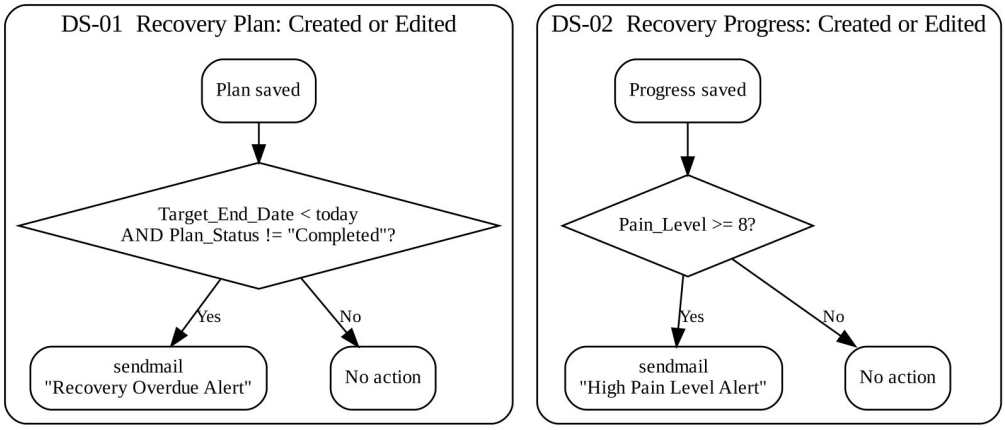


Figure 4.4: Decision Logic of the Deluge Alert Scripts DS-01 and DS-02

4.5 User Interface and Navigation Design

The application uses Zoho Creator's left-hand navigation, with one section per form (each holding the form and its report) and the dashboard page last. Figure 4.5 shows the navigation structure and the wireframe of the Sports_Recovery_Dashboard page planned before it was built in the page builder.

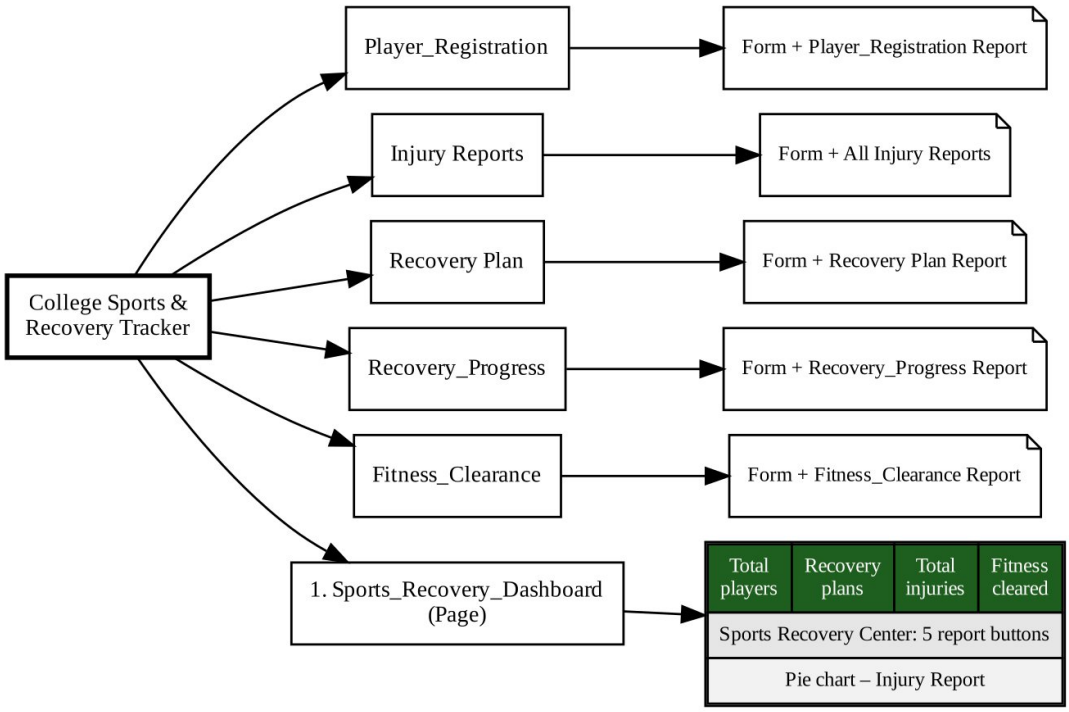


Figure 4.5: Navigation Structure and Wireframe of the Sports_Recovery_Dashboard Page

4.6 Access Control Design

Table 4.7 defines the intended permission matrix (A = Add, E = Edit, V = View, D = Delete, — = No access). In the current build only the Administrator role is in use; the Physiotherapist profile is the planned next step (see Sections 5.8 and 9.2).

Table 4.7: Role–Permission Matrix

Component	Sports Coordinator / Administrator	Physiotherapist
Form: Player_Registration	A E V D	V
Form: Injury Report	A E V D	V E
Form: Recovery Plan (with subform)	A E V D	A E V
Form: Recovery_Progress	A E V D	A E V
Form: Fitness_Clearance	A E V D	A V
Page: Sports_Recovery_Dashboard	V	V

CHAPTER 5
IMPLEMENTATION

5.1 Application Setup

The application was created in the Zoho Creator workspace of the developer account (kaileshm65552) on the zoho.in data centre under a Free Trial licence, with the name **College Sports & Recovery Tracker** and the link name college-sports-recovery-tracker. It was built from scratch using the Design, Workflow and Settings tabs of the application builder.

The navigation was organised into six sections that follow the injury lifecycle: Player_Registration, Injury Reports, Recovery Plan, Recovery_Progress, Fitness_Clearance and "1. Sports_Recovery_Dashboard". Each form section contains the form and its report (for example, Player_Registration and Player_Registration Report). The application uses the platform's dark-sidebar theme. Figure 5.1 shows the navigation with the Player_Registration form open.

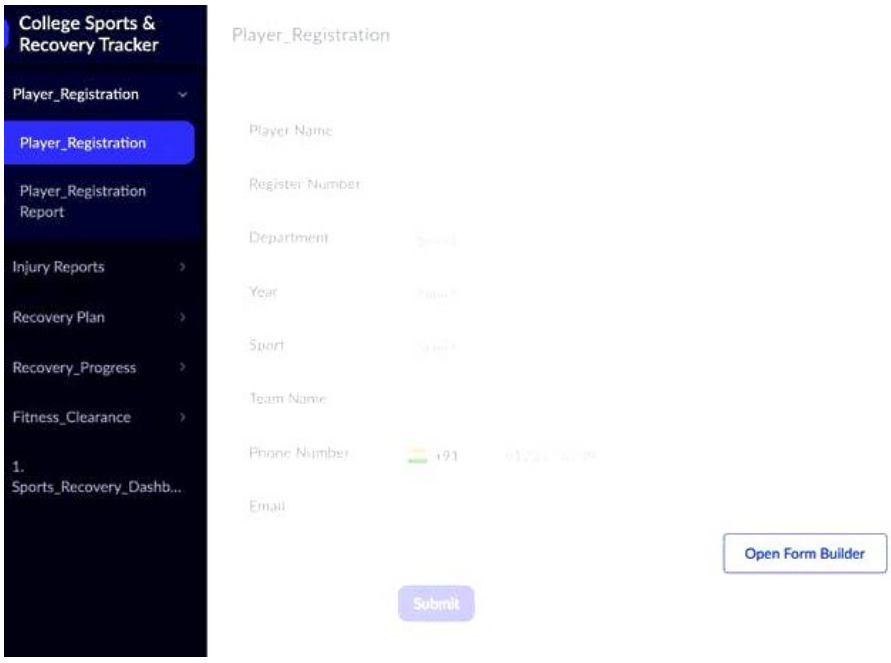


Figure 5.1: Application Navigation Sections with the Player_Registration Form

5.2 Forms and Data Capture

Five forms were built in the form builder, one per entity of the data model, using twelve distinct field types — Auto Number, Single Line, Multi Line, Number, Email, Phone, Date, Drop Down, Lookup, Image, Decision Box and Subform. Every form starts with

an Auto Number ID field (start index 1) so that each record has a readable, unique identifier.

Player_Registration form

Registers each player (Figure 5.2) with Player ID, Player Name, Register Number, Department, Year and Sport (drop-downs), Team Name, Phone Number (with +91 country code) and Email. It is the parent of Injury Report.

Injury Report form

Records an injury (Figure 5.3). The Player_Registration lookup links the injury to a registered player; Sport, Body Part, Injury Type, Severity, Cause and Injury Status are drop-downs, so injuries are classified consistently; Description is a Multi Line field and Injury Photo is an Image field for a visual record.

Recovery Plan form

Defines the treatment for an injury (Figure 5.4). It links to Injury Report through a lookup, records Start Date, Target End Date, Physiotherapist Name and Plan Status, and contains the **Recovery_Activities subform**, in which each row is one exercise with Activity Name, Duration in Minutes, Sessions per Week, Instructions and Activity Status. The subform keeps the activities inside their plan instead of in a separate, loosely linked form.

Recovery_Progress form

Logs each follow-up (Figure 5.5): a lookup to the Recovery Plan, Progress Date, Pain Level and Progress Percentage (Number fields), Exercise Completed (Decision Box), Physiotherapist Remarks and Current Status.

Fitness_Clearance form

Documents the return-to-play decision (Figure 5.6): a lookup to the Injury Report, Evaluation Date, Fitness Result, Recommendation, Reviewed By, Approval Status and Remarks. Figure 5.7 shows how the Player_Registration form is laid out in the Design builder’s web preview.

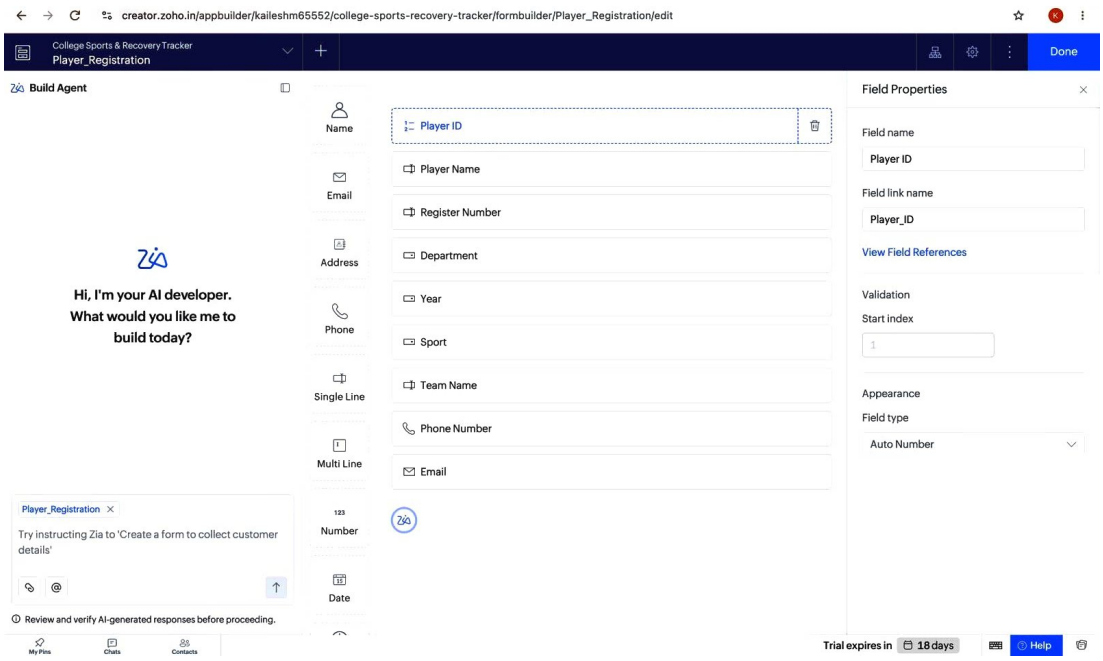


Figure 5.2: Player_Registration Form in the Form Builder

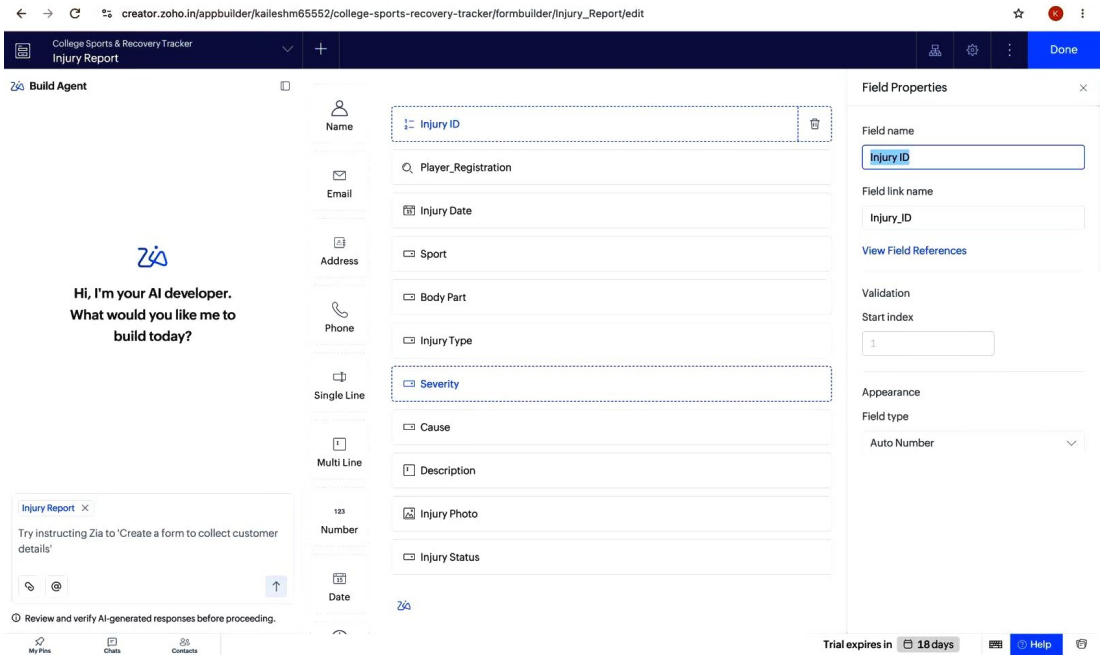


Figure 5.3: Injury Report Form in the Form Builder

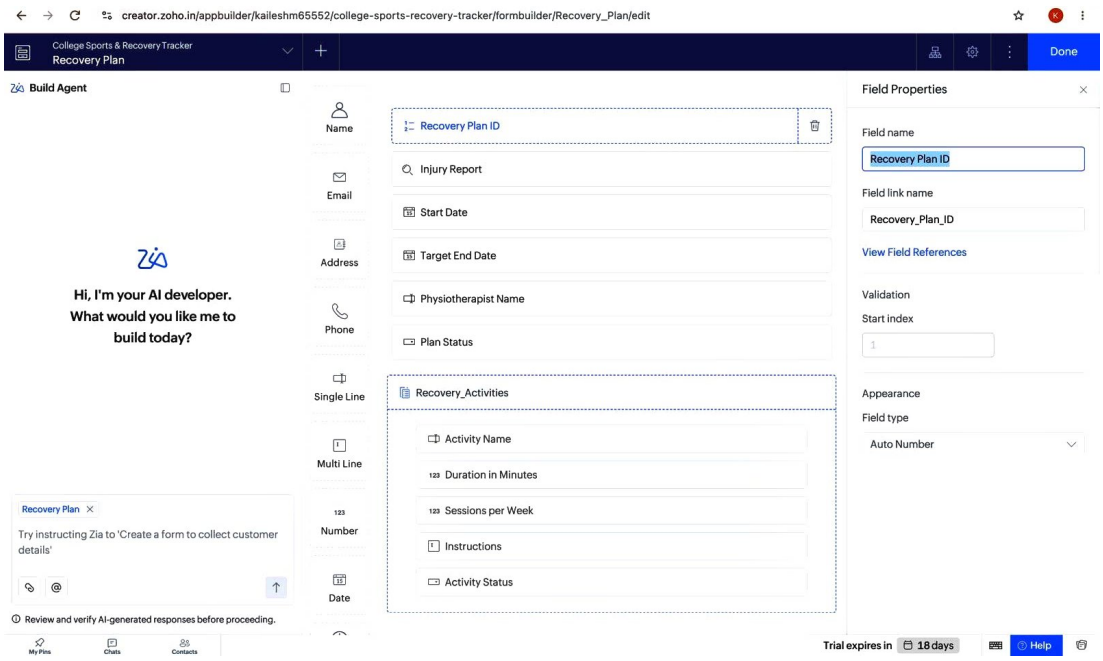


Figure 5.4: Recovery Plan Form with the Recovery_Activities Subform

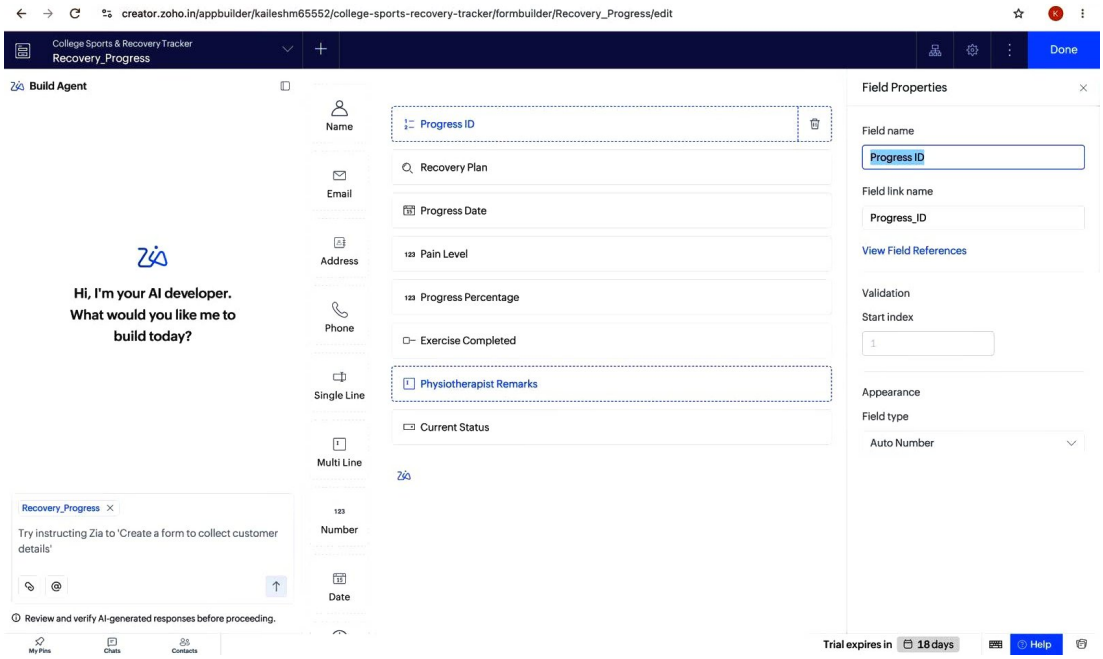


Figure 5.5: Recovery_Progress Form in the Form Builder

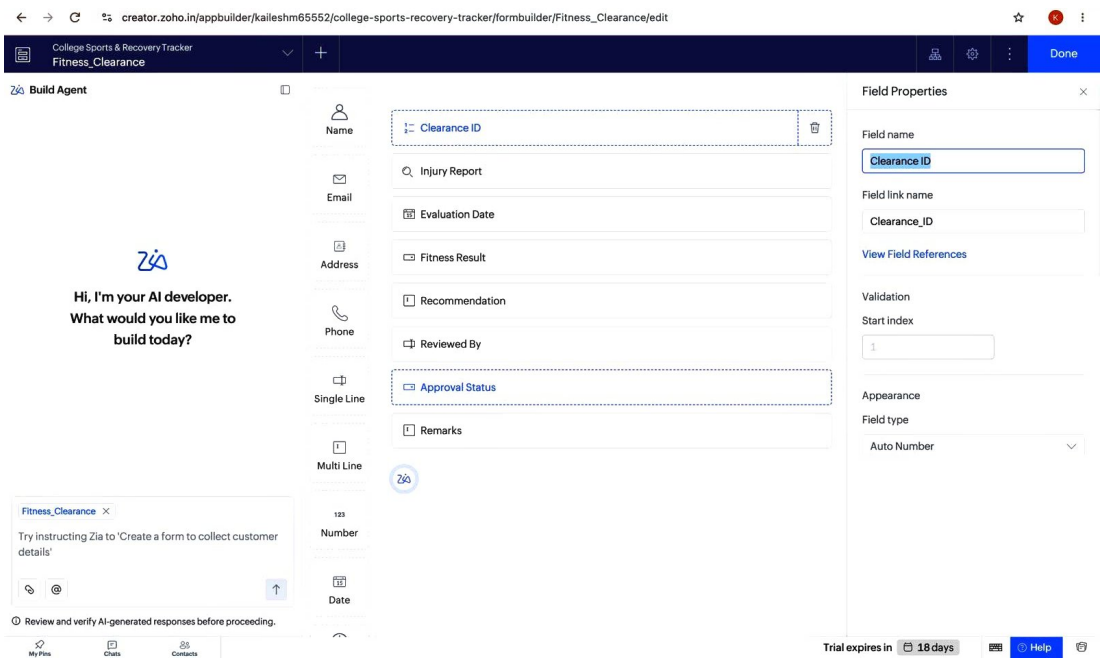


Figure 5.6: Fitness_Clearance Form in the Form Builder

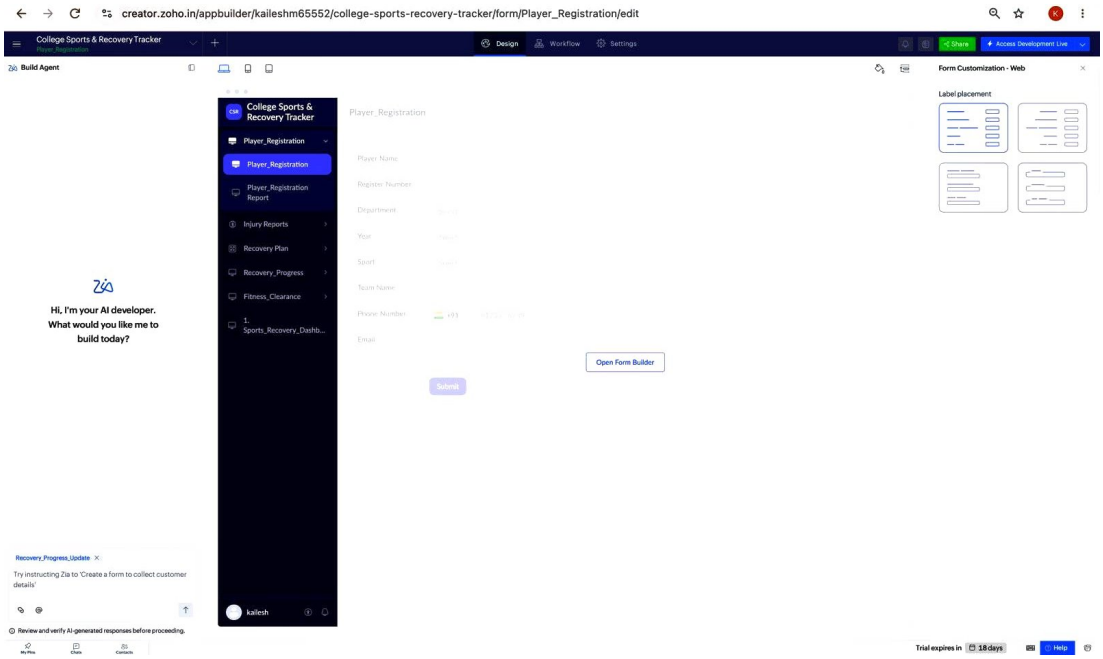


Figure 5.7: Player_Registration Form Layout in the Design Builder (Web Preview)

5.3 Reports

A list report was generated for each form, and a pie chart report was created on Injury Report for the dashboard. The reports are summarised in Table 5.1.

Table 5.1: Reports Implemented

Report Name	Type	Source Form	Filter / Grouping / Sorting	Custom Actions	Roles
Player_Registration Report	List	Player_Registration	All records	Default view / edit / delete	Administrator
All Injury Reports	List	Injury Report	All records	Default view / edit / delete	Administrator
Recovery Plan Report	List	Recovery Plan	All records (with subform rows in detail view)	Default view / edit / delete	Administrator
Recovery_Progress Report	List	Recovery_Progress	All records	Default view / edit / delete	Administrator
Fitness_Clearance Report	List	Fitness_Clearance	All records	Default view / edit / delete	Administrator
Pie – Injury Report	Chart (Pie)	Injury Report	Grouped by	—	Administrator

All five list reports are reachable from the dashboard's "Sports Recovery Center" buttons (Figure 5.8), so a user can jump from the overview straight to the detailed records. Grouping, conditional formatting, custom action buttons and Kanban or Calendar report types were not configured in this build; a Kanban report on Plan Status and a Calendar report on Target End Date are listed as enhancements in Section 9.2.



Figure 5.8: Report Navigation Buttons in the Sports Recovery Center Panel

5.4 Pages and Dashboards

The **Sports_Recovery_Dashboard** page was built in the page builder as the landing view for the coordinator (Figure 5.9). It has three parts. The top row contains four KPI panels that use count functions on the forms: **Total players registered**, **Recovery plans**, **Total injuries** and **Fitness cleared players**. The middle section, titled "Sports Recovery Center", contains five buttons that open the Player_Registration Report, All Injury Reports, Recovery Plan Report, Recovery_Progress Report and Fitness_Clearance Report. The bottom panel, "Sports_recovery_dashboard", embeds the Pie – Injury Report chart. The page uses a consistent green theme, and because every panel reads live form data it updates as soon as a record is added.

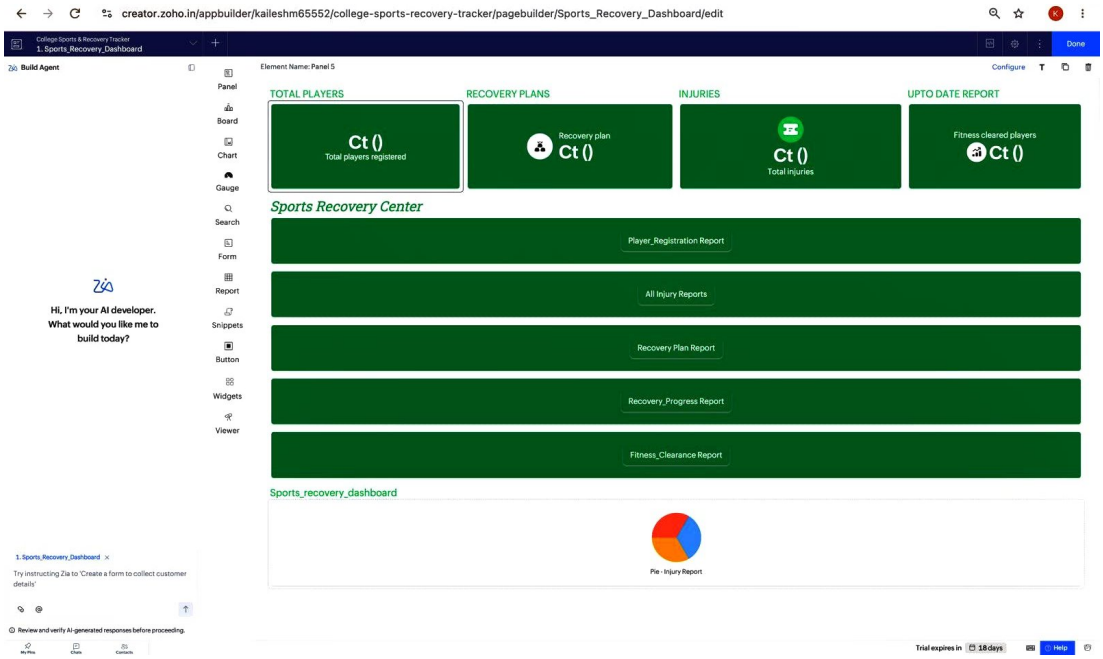


Figure 5.9: Sports_Recovery_Dashboard Page in the Page Builder

5.5 Deluge Scripting

Table 5.2: Deluge Scripts Implemented

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-01	Recovery Plan → Created or Edited → Successful form submission (workflow "Recovery_Overdue_Alert")	E-mail the coordinator when a plan is past its target end date and not completed	if, date comparison with zoho.currentdate, logical AND, sendmail

Script ID	Location / Trigger	Purpose	Deluge Concepts Used
DS-02	Recovery_Progress → Created or Edited → Successful form submission (workflow "Recovery_Progress_Update")	E-mail a high-pain alert when Pain Level is 8 or more	if, numeric comparison, sendmail with HTML message

Code 5.1: DS-01 Recovery Overdue Alert (Recovery Plan → Created or Edited)

```
// DS-01 | Form: Recovery_Plan | Event: Created or Edited ->
Successful form submission
if(Target_End_Date < zoho.currentdate && Plan_Status != "Completed")
{
    sendmail
    [
        from :"gowthamraj.m.2024.cse@rajalakshmi.edu.in"
        to :"kaileshm65552@gmail.com"
        subject :"Recovery Overdue Alert"
        message :"A recovery plan has passed its target end date and
is not yet completed."
    ]
}
```

Code 5.2: DS-02 High Pain Level Alert (Recovery_Progress → Created or Edited)

```
// DS-02 | Form: Recovery_Progress | Event: Created or Edited ->
Successful form submission
if(Pain_Level >= 8)
{
    sendmail
    [
        from :"gowthamraj.m.2024.cse@rajalakshmi.edu.in"
        to :"kaileshm65552@gmail.com"
        subject :"High Pain Level Alert"
        message :"<div><span>A recovery progress entry has recorded a
pain level of 8 or above.</span></div>"
    ]
}
```

DS-01 runs each time a Recovery Plan is created or edited. It compares the plan's Target_End_Date with zoho.currentdate (today's date) and checks that Plan_Status is not "Completed". Only when both conditions are true does the sendmail task e-mail a "Recovery Overdue Alert" to the coordinator. A plan that is on schedule, or already completed, produces no e-mail.

DS-02 runs each time a Recovery_Progress entry is created or edited and sends a "High Pain Level Alert" when Pain_Level is 8 or more on the 0–10 scale. The message body uses simple HTML, which sendmail renders in the e-mail. Both scripts are short by design: the decision logic is a single condition, and everything else (record storage,

linking, dashboards) is handled by configuration. Figures 5.10 and 5.11 show the scripts in the workflow builder; the complete code is in Appendix C.

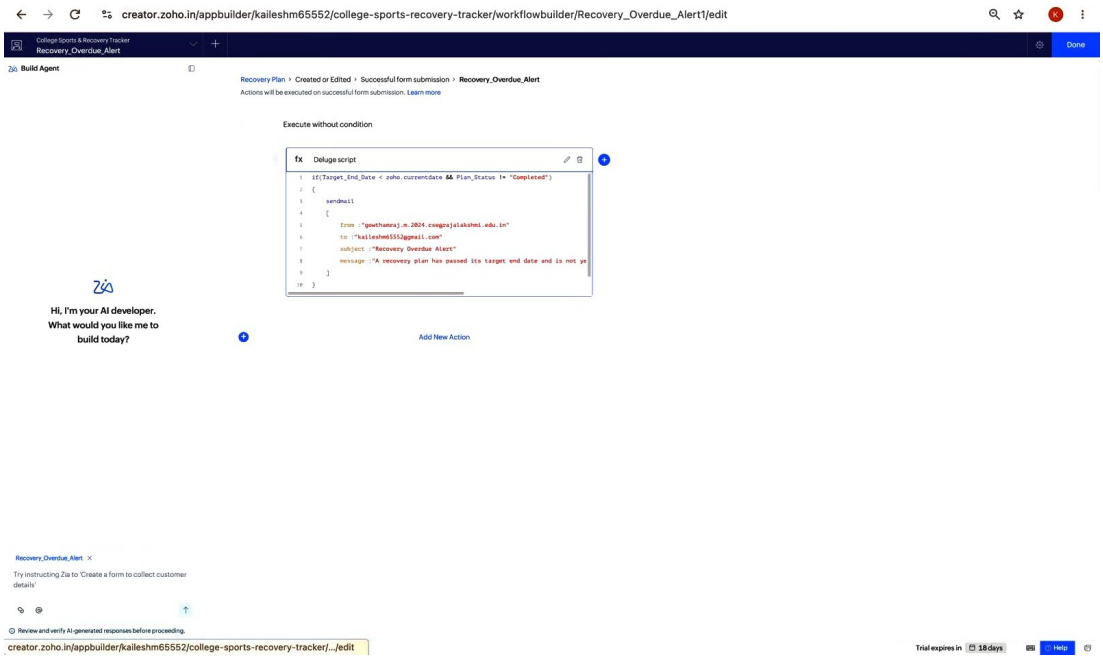


Figure 5.10: DS-01 in the Recovery_Overdue_Alert Workflow

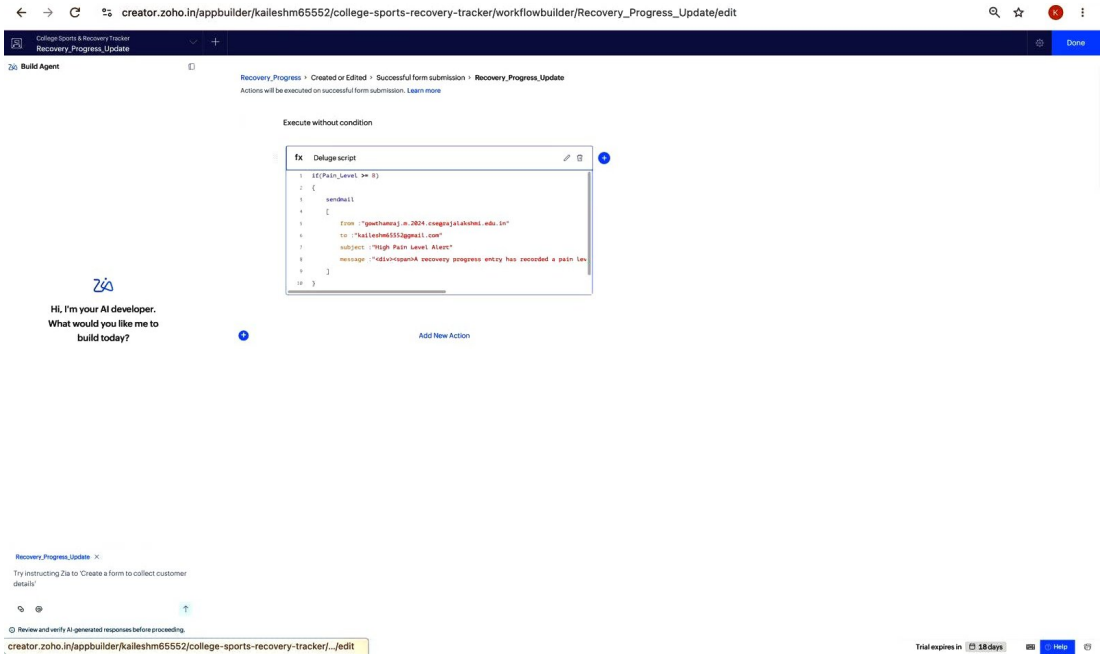


Figure 5.11: DS-02 in the Recovery_Progress_Update Workflow

5.6 Workflows and Automation

Three form workflows were configured, all created on 20 October 2026 and enabled (Table 5.3). Player_Registration and Fitness_Clearance appear under “Forms without workflows”.

Table 5.3: Workflows Implemented

Workflow Name	Type	Trigger / Event	Condition	Action(s)	Script Ref.
Injury_Reported_Alert	Form Workflow (Injury Report)	Created → Successful form submission	Execute without condition	Send Mail to kaileshm65552@gmail.com	—
Recovery_Overdue_Alert	Form Workflow (Recovery Plan)	Created or Edited → Successful form submission	Inside script: Target_End_Date < today AND Plan_Status != Completed	Deluge sendmail "Recovery Overdue Alert"	DS-01
Recovery_Progress_Update	Form Workflow (Recovery_Progress)	Created or Edited → Successful form submission	Inside script: Pain_Level >= 8	Deluge sendmail "High Pain Level Alert"	DS-02

Injury_Reported_Alert is entirely no-code: a Send Mail action fires on every new injury report (Figure 5.12). The other two workflows run on both creation and edit, so changing a plan's status or a progress entry re-evaluates the alert condition. Figures 5.13 to 5.15 show the workflow list for each form. Scheduled workflows, approval processes, Blueprints and batch workflows were not used; Section 8.4 explains why the overdue alert would be more reliable as a scheduled workflow.

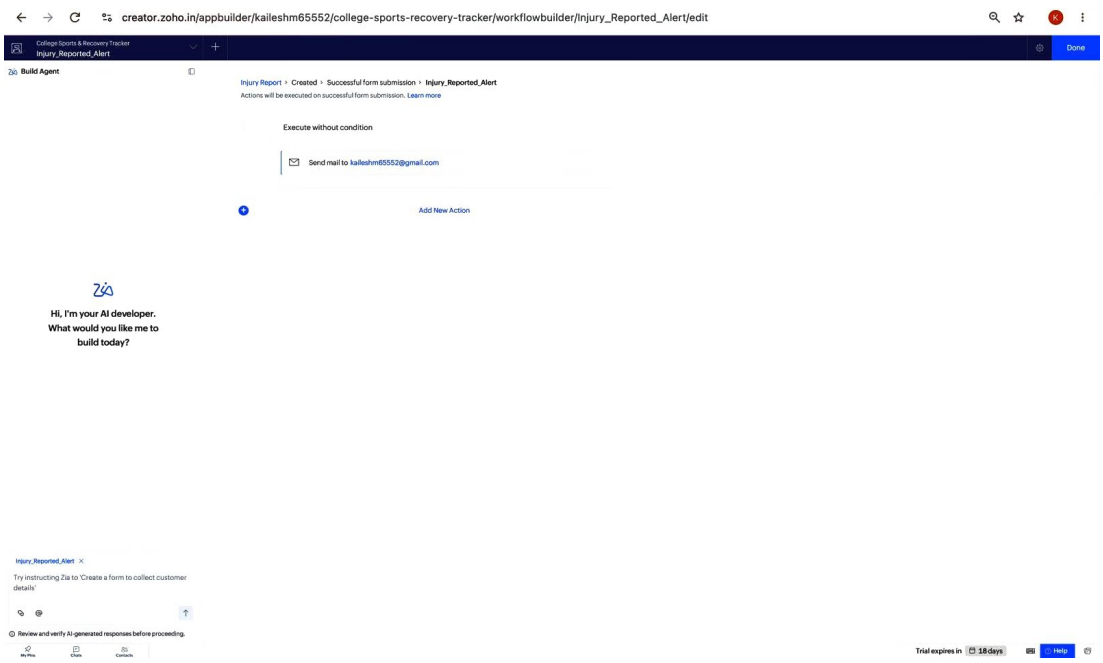


Figure 5.12: Injury_Reported_Alert Workflow (Send Mail Action)

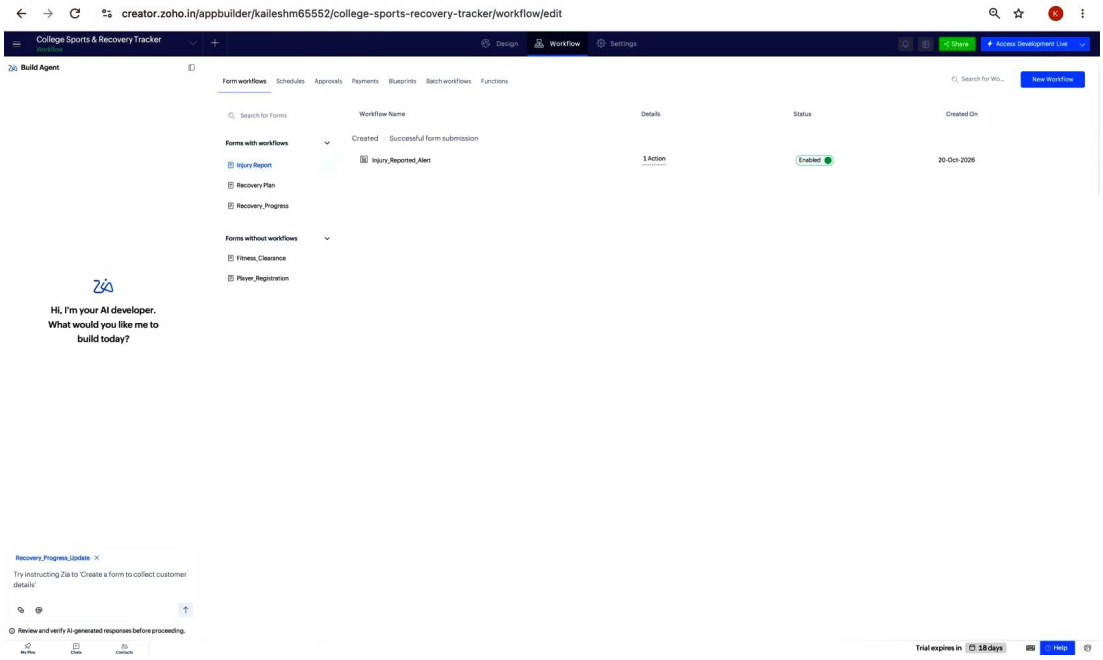


Figure 5.13: Form Workflow List – Injury Report

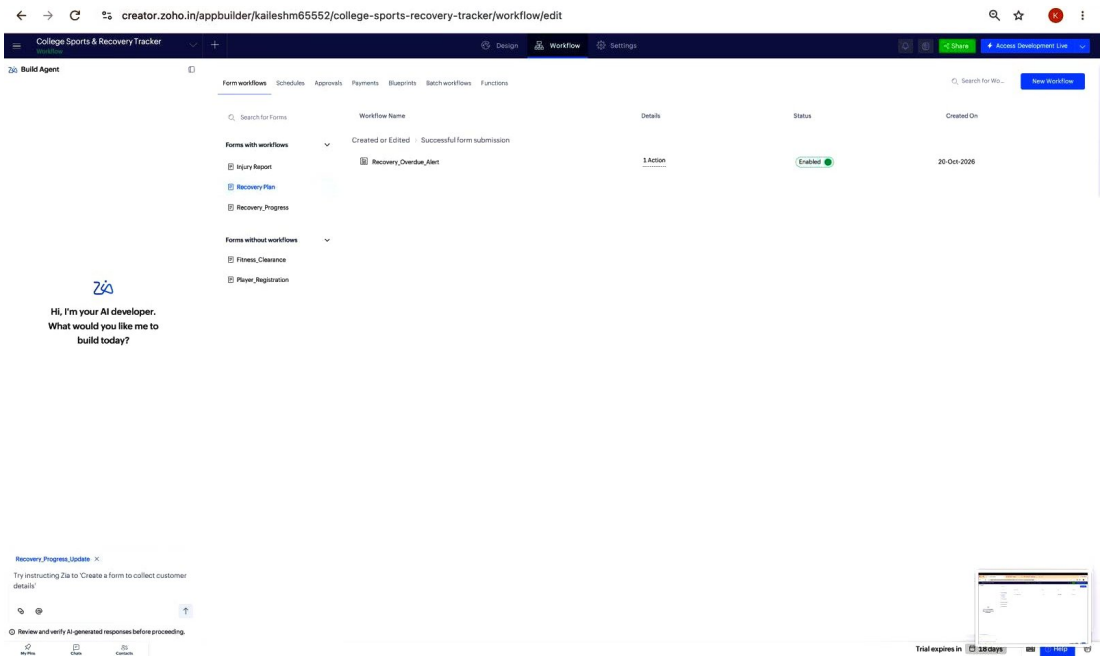


Figure 5.14: Form Workflow List – Recovery Plan

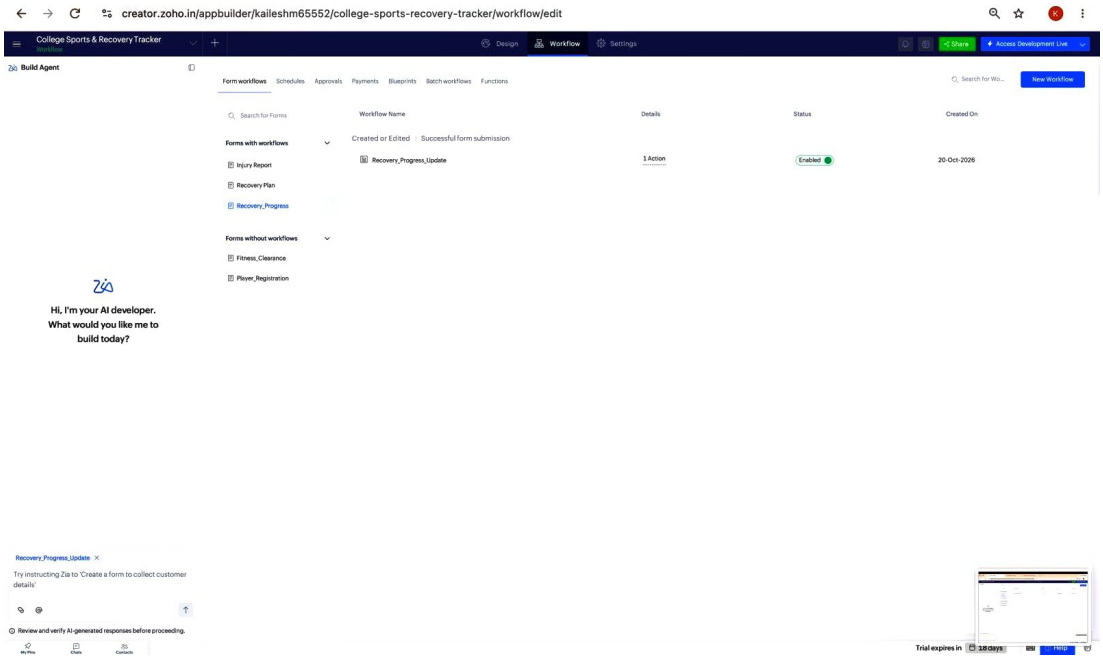


Figure 5.15: Form Workflow List – Recovery_Progress

5.7 Integrations

The application uses Zoho's own mail service in two ways and connects to no external system (Table 5.4).

Table 5.4: Integrations Implemented

Integration	Service / Endpoint	Method	Authentication	Data Exchanged / Purpose
Injury alert	Zoho mail service (built-in)	Send Mail action in Injury_Reported_Alert	Zoho account of the application owner	Notify the coordinator of every new injury
Overdue / high-pain alerts	Zoho mail service (built-in)	Deluge sendmail task in DS-01 and DS-02	Zoho account of the application owner	Notify the coordinator of overdue plans and pain level ≥ 8

Justification. The injury-management process is internal to the college, so the core requirements could be met without external data. No connection, invokeurl REST call, webhook or Creator REST API was configured, and no Postman testing was needed. Useful future integrations include Zoho Cliq or WhatsApp Business (through a connection) for instant alerts to coaches at the ground, Zoho Analytics for season-wise injury analysis, and Zoho Bookings for physiotherapy appointments.

5.8 User Management and Role-Based Access

The application is currently operated through the application owner's account ("kailesh"), which has the Administrator role and full access to all forms, reports and the dashboard. Additional users, custom roles and permission profiles were not yet configured, so the Physiotherapist column of Table 4.7 is a design target.

Implementing it requires three steps in Settings → Users and Control: (1) create a "Physiotherapist" permission profile with Add, Edit and View on Recovery Plan and Recovery_Progress, Add and View on Fitness_Clearance, and View on Player_Registration, Injury Report and the dashboard; (2) restrict Delete to the Administrator; and (3) invite the physiotherapist and assign the profile. Because injury records are health-related, restricting access in this way is the first enhancement listed in Section 9.2.

5.9 AI and Analytics Features

Analytics. Analytics are delivered on the Sports_Recovery_Dashboard (Figure 5.9): four KPI panels compute live counts of players, recovery plans, injuries and fitness-cleared players, and a pie chart aggregates injury reports. Zoho Analytics and pivot reports were not used.

AI features. The Zoho Creator builder includes the Zia Build Agent ("Hi, I'm your AI developer. What would you like me to build today?"), shown in Figure 5.16, which can generate forms, fields and scripts from natural-language prompts. No Zia AI fields such as prediction, OCR, keyword extraction or sentiment analysis were added.

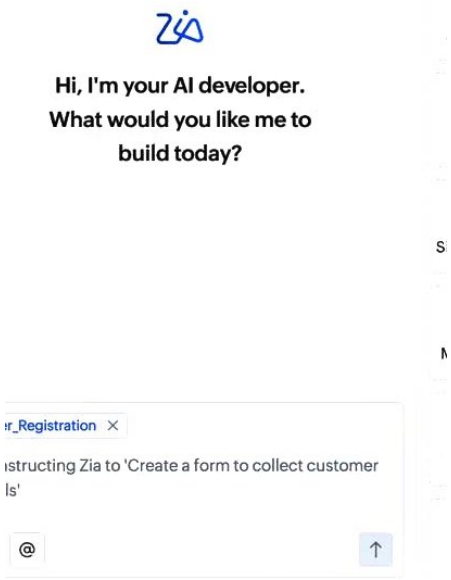


Figure 5.16: Zia Build Agent (AI Developer) Panel Available in the Application Builder

5.10 Pro-Code Extension

No pro-code extension was built. All requirements of the current scope — data capture, relationships, nested activities, alerts and the dashboard — could be met with Zoho Creator's low-code features, and no custom Deluge functions, Zoho Catalyst functions, JavaScript widgets or Creator REST API clients were required.

A pro-code extension would become justified for capabilities the platform does not provide natively — for example, a Catalyst function that analyses injury photos or produces season-wise injury-risk scores, a custom widget that draws a body-map showing where injuries occur most often, or a REST API client that pushes clearance status to a team-selection system.

5.11 Mobile and Portal Access

The application was built and tested in the web interface. Zoho Creator makes every application available on its Android and iOS mobile app without separate development, which is particularly useful for logging injuries at the sports ground; the Design builder provides desktop, tablet and mobile layout previews (visible at the top of Figure 5.7). Mobile-specific layouts, offline mode and a portal for players were not configured in this build.

CHAPTER 6
TESTING AND DEBUGGING

6.1 Testing Strategy

Testing was carried out in the Development environment at four levels. **Unit testing** exercised each form on its own: valid records were submitted to confirm that the Auto Number ID is generated, lookups list the correct parent records, typed fields reject invalid input (for example, text in a Number field or a malformed e-mail) and the Recovery_Activities subform accepts multiple rows. **Script testing** targeted the alert boundaries: a plan with a past target date and status other than Completed, the same plan with status Completed, a plan with a future target date, and progress entries with pain levels 7, 8 and 10. **Workflow testing** confirmed that each alert e-mail arrives once per qualifying save. **Dashboard testing** checked that each KPI count and the pie chart change when records are added.

Role-based access testing was not applicable because only the Administrator role exists in the current build. The test cases, traced to the functional requirements of Table 3.2, are listed in Table 6.1.

6.2 Test Cases

Table 6.1: Test Cases

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-01	FR-01	Register a player	Valid player details	Record saved; Player ID auto-generated	As expected	Pass
TC-02	FR-01	Enter an invalid e-mail	“player@”	Invalid e-mail error; not saved	As expected	Pass
TC-03	FR-02	Report an injury with photo	Select player, fill drop-downs, upload image	Saved with Injury ID; photo stored	As expected	Pass
TC-04	FR-02	Lookup lists registered players only	Open Player_Registration lookup	Only existing players shown	As expected	Pass
TC-05	FR-03	Injury alert e-mail	Submit a new injury	E-mail received by coordinator	As expected	Pass
TC-06	FR-04	Create plan with three activities	Three subform rows	Plan saved with all three rows	As expected	Pass
TC-07	FR-04	Text in Duration in Minutes	“thirty”	Number field rejects input	As expected	Pass

TC ID	FR Ref.	Test Description	Input / Steps	Expected Result	Actual Result	Status
TC-08	FR-05	Overdue plan, not completed	Target End Date = yesterday; status In Progress	"Recovery Overdue Alert" e-mailed	As expected	Pass
TC-09	FR-05	Overdue plan, completed	Target End Date = yesterday; status Completed	No e-mail	As expected	Pass
TC-10	FR-05	Plan within schedule	Target End Date = next week	No e-mail	As expected	Pass
TC-11	FR-06	Log progress entry	Pain 4, Progress 60%, exercise ticked	Saved with Progress ID	As expected	Pass
TC-12	FR-07	Pain level at threshold	Pain Level = 8	"High Pain Level Alert" e-mailed	As expected	Pass
TC-13	FR-07	Pain level below threshold	Pain Level = 7	No e-mail	As expected	Pass
TC-14	FR-08	Record fitness clearance	Result Fit; Approval Approved	Saved with Clearance ID	As expected	Pass
TC-15	FR-09	KPI counts update	Add one player and one injury	Total players and Total injuries increase by one	As expected	Pass
TC-16	FR-09	Dashboard report buttons	Click each of the five buttons	Corresponding report opens	As expected	Pass
TC-17	FR-10	Edit and delete from a report	Edit an injury status; delete a test record	Changes reflected in report	As expected	Pass
TC-18	NFR-02	Open application while signed out	Private browser window	Zoho sign-in page shown	As expected	Pass

6.3 Debugging Deluge Scripts

The alert scripts were debugged mainly through their output. Each script was tested at and around its threshold (TC-08 to TC-13), and the arrival or absence of the alert e-mail confirmed which branch had executed. The Deluge editor's save-time syntax check caught errors in field link names, which must match the form exactly (for example, Target_End_Date and Plan_Status). Where a condition did not behave as expected, a temporary info statement printing the field values was added and the result viewed in the workflow's execution log before removing it.

One real issue concerned the comparison value for the plan status: the condition must compare with the exact drop-down choice text (“Completed”), including capitalisation, otherwise completed plans would still trigger the overdue alert. The value was checked against the Plan_Status field’s choices. Figure 6.1 should show an alert e-mail received during testing.

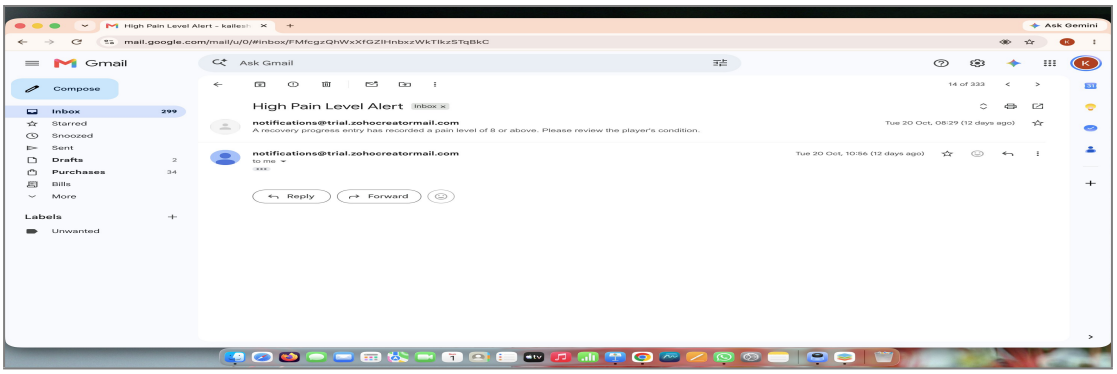


Figure 6.1: Alert E-mail Received During Testing

6.4 Defect Log

Table 6.2: Defect Log

Bug ID	Description	Root Cause	Fix Applied	Status
BUG-01	Completed plans could still trigger the overdue alert	Status comparison must match the drop-down choice text exactly	Compared with the exact value “Completed” used in Plan_Status	Closed
BUG-02	Overdue alert does not fire for plans that silently pass their end date	DS-01 runs only when a plan is created or edited, not daily	Move the check to a daily scheduled workflow over all open plans	Open
BUG-03	Records can be saved with key fields empty (e.g., no player or injury date)	Mandatory flag not enabled on the forms	Enable Mandatory on lookups, dates, Severity, Pain Level and status fields	Open
BUG-04	Alert recipient and sender addresses are hard-coded	E-mail addresses written directly in the workflow and scripts	Use zoho.adminuserid as sender and read the recipient from a settings record	Open

6.5 User Acceptance Testing

Table 6.3: User Acceptance Feedback

User	Role Tested	Feedback	Action Taken
Haripraveen N	Student	The dashboard is easy to use and helps monitor player recovery status.	Improved dashboard navigation and recovery information display.
Gokulan V	Student	The application is easy to use. It helps track workout activities, injuries, and recovery progress.	Improved the user interface and made recovery progress easier to understand.

CHAPTER 7
DEPLOYMENT, SECURITY AND GOVERNANCE

7.1 Environments and Deployment Strategy

Zoho Creator provides Development, Stage and Production environments so that changes can be built and tested without affecting live users. All work on this application, including the screenshots in this report, was done in the **Development** environment, indicated by the "Access Development Live" button in the builder toolbar (Figure 7.1).

The application has not yet been published to Stage or Production. The planned release path is: (1) freeze the Development version and record release notes; (2) publish to Stage and repeat the test cases in Table 6.1 with a second user; (3) remove test records, enable mandatory fields and configure the Physiotherapist profile; (4) publish to Production and verify the live URL; and (5) keep the previous version available for rollback. Table 7.1 shows the status of each checklist item.



Figure 7.1: Application Builder Toolbar Showing the Development Environment (Access Development Live)

Table 7.1: Deployment Checklist

S.No.	Checklist Item	Done (✓)
1	All test cases executed in Development / Stage	✓ (Development)
2	Sample/test data removed or isolated	✗
3	Roles, permissions and users configured in Production	✗
4	Connections authorised in Production	N/A (no connections used)
5	Schedules and workflows enabled	✓ (all 3 form workflows enabled)
6	Application published and URL verified	✗ (Development URL only)
7	Version / release notes recorded	✗

7.2 Performance Optimisation

With a small number of records, performance was not a constraint and no formal before-and-after timings were recorded. The design nevertheless follows the Unit IV optimisation practices so that it keeps performing as data accumulates over seasons (Table 7.2).

Table 7.2: Performance Optimisation Results

Optimisation Applied	Component	Before	After	Improvement
Alert scripts evaluate only the current record's fields — no fetch or loop over other records	DS-01, DS-02	Not applicable (designed in)	Constant work per save	Cost independent of the number of plans or progress entries
Recovery activities stored in a subform instead of a separate form with lookups	Recovery Plan	—	—	Plan and activities load together; no extra lookup queries
Dashboard KPIs computed by platform count functions instead of manual tallies	Sports_Recovery_Dashboard	Manual counting	Live counts	No manual reporting step

7.3 Security Features

Security in the current build relies mainly on platform controls. Because the application holds health-related information about players, the planned controls in Table 7.3 are important before production use.

Table 7.3: Security Controls Implemented

Security Control	Where Applied	Threat Mitigated
Zoho account authentication	Entire application	Unauthenticated access to player and injury data
Application sharing limited to the owner account	Application settings	Unauthorised viewing or editing of records
Lookups and typed fields	All five forms	Orphan records and malformed data
Planned: Physiotherapist profile without Delete rights	All forms	Accidental or malicious deletion of medical records

Security Control	Where Applied	Threat Mitigated
Planned: field-level hiding of Injury Photo and remarks for non-medical users	Injury Report, Recovery_Progress	Exposure of sensitive health information

7.4 Governance and Administration

Governance practices were kept simple. **Naming conventions:** forms, fields and workflows have descriptive names whose link names (for example, Target_End_Date, Pain_Level, Recovery_Overdue_Alert) are self-explanatory in Deluge. **Identifiers:** every form has an Auto Number ID, giving each record a stable reference for discussion and audit. **Change history:** each workflow records its creation date (20-Oct-2026) and enabled status in the Workflow tab. **Ownership:** the application owner account administers the app; when responsibility passes to new staff, they should be added as administrators rather than sharing credentials.

Audit trail, scheduled backups and user-licence management were not configured. Before production use, the audit trail should be enabled on Injury Report, Recovery_Progress and Fitness_Clearance so that every change to health-related data is traceable, and a periodic export of all reports should be scheduled as a backup.

CHAPTER 8
RESULTS AND DISCUSSION

8.1 Application Output

The working application is demonstrated through one realistic scenario: a football player sprains an ankle during practice. **Step 1 (Figure 8.1):** the coordinator records the injury in the Injury Report form — selecting the player, setting Body Part "Ankle", Injury Type "Sprain" and Severity, and uploading a photo — and an Injury_Reported_Alert e-mail is sent. **Step 2 (Figure 8.2):** the physiotherapist's recovery plan is entered with a target end date and three activities in the Recovery_Activities subform. **Step 3:** progress entries are logged over the following weeks; one entry with Pain Level 8 triggers a High Pain Level Alert, prompting the physiotherapist to adjust the plan. **Step 4:** when the player recovers, a Fitness_Clearance record is entered with result "Fit" and approval granted. **Step 5 (Figure 8.3):** the dashboard KPIs reflect the injury, the plan and the newly cleared player.

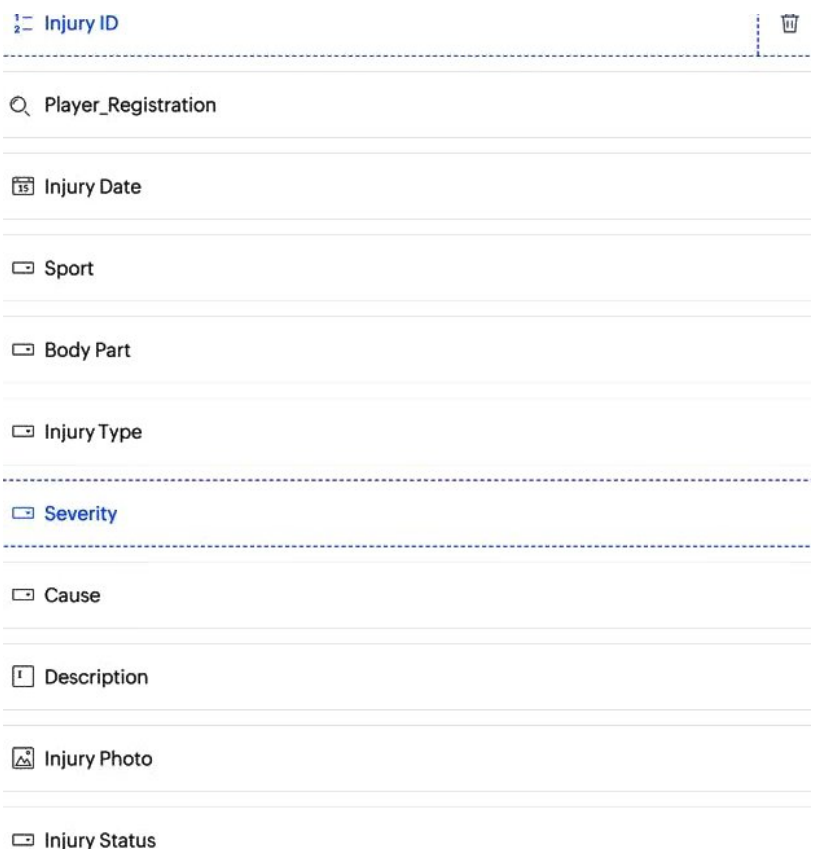


Figure 8.1: Scenario Step 1 – Injury Recorded in the Injury Report Form

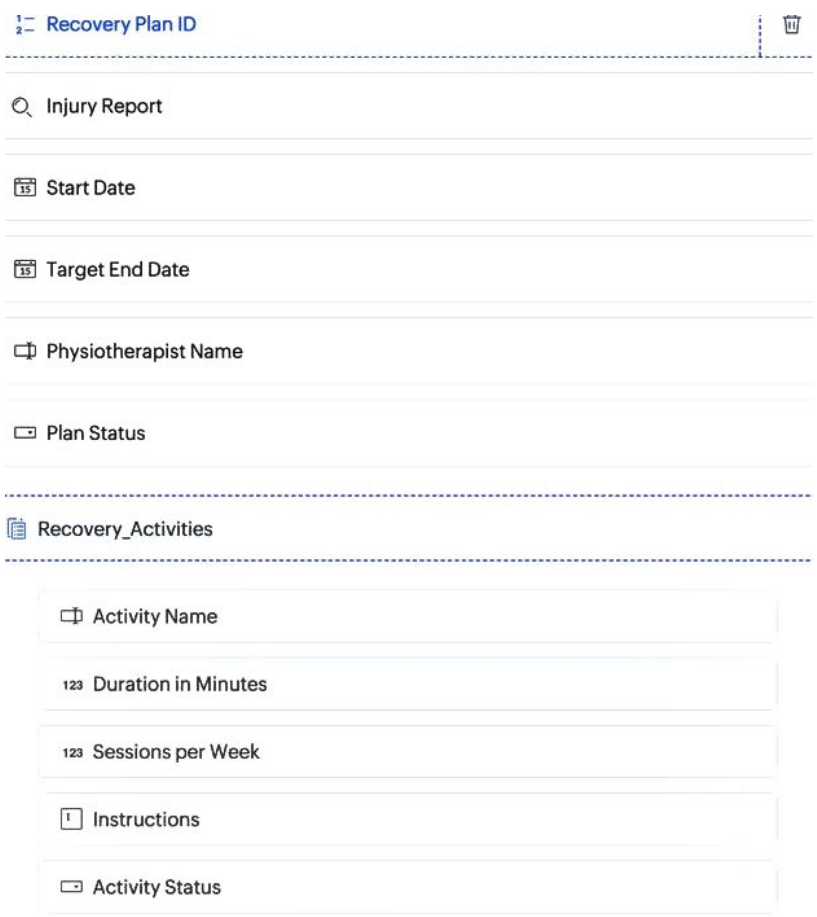


Figure 8.2: Scenario Step 2 – Recovery Plan with Activities in the Subform



Figure 8.3: Scenario Step 5 – Dashboard KPI Panels Reflect Players, Plans, Injuries and Clearances

8.2 Achievement of Objectives

Table 8.1: Mapping of Objectives to Outcomes

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Relational data model with five forms, auto IDs and subform	Sections 4.2, 4.3, 5.2	Five forms, four lookups, Recovery_Activities subform, 12 field types (Figures 5.2–5.6)
Automatic injury alert	Section 5.6	Send Mail on every new injury (TC-05, Figure 5.12)
Overdue and high-pain detection	Sections 5.5, 5.6 (DS-01, DS-02)	Alerts at the correct boundaries (TC-08 to TC-13, Figures 5.10, 5.11)

Objective (Sec. 1.3)	Implemented Through	Outcome / Evidence
Dashboard with KPIs, report links and chart	Section 5.4	Four live counts, five report buttons and pie chart (Figure 5.9, TC-15, TC-16)
Systematic testing and deployment path	Chapters 6 and 7	18 test cases executed in Development; production deployment planned (Table 7.1)

8.3 Comparison with the Existing System

Table 8.2: Existing System versus Proposed Application

Parameter	Existing System	Proposed Application	Improvement
Recording an injury	Verbal report, notebook or chat message	Structured Injury Report with classification and photo	Consistent, searchable record
Recovery plan	Oral or paper instructions	Plan with activities, durations and sessions in a subform	Nothing lost or misremembered
Detecting problems	Only if someone asks	Automatic overdue and high-pain e-mails	Proactive instead of reactive
Return-to-play decision	Verbal, undocumented	Fitness_Clearance with reviewer and approval status	Documented and accountable
Season overview	Manual compilation, often impossible	Live dashboard KPIs and injury chart	Real-time visibility

8.4 Limitations of the Low-Code Approach

Building this application exposed several practical limits of the low-code approach. **Event-driven versus time-driven logic:** form workflows run only when a record is saved, so DS-01 cannot notice a plan that quietly passes its target date without anyone editing it (BUG-02); a proper overdue check needs a scheduled workflow, which is a different construct on the platform. **UI customisation:** the dashboard could be themed with colours and panels, but layout options are limited to the page builder's components, and a richer visual such as a body-map of injuries would need a custom widget.

Hard-coded values: alert addresses and the pain threshold are written directly into the workflows, which is quick to build but must be edited in several places if they change (BUG-04). **Licensing:** the Free Trial limits users and duration. **Vendor lock-in:** forms, workflows and Deluge scripts are specific to Zoho Creator; the data can be exported, but the logic would have to be rebuilt on another platform. For a college sports department these trade-offs are acceptable given the speed of building and ease of maintenance.

8.5 Reflection on the No-Code → Low-Code → Pro-Code Spectrum

Most of the application sits at the **no-code** end of the spectrum: all five forms, the subform, lookups, Auto Number IDs, image upload, the list reports, the pie chart, the dashboard with its KPI counts and buttons, and the Injury_Reported_Alert e-mail were configured entirely through visual builders. Only the two conditional alerts required **low-code** Deluge, because each needs a condition on field values before sending mail. Nothing in the current scope required **pro-code**.

The move to pro-code would come with the more ambitious enhancements — image analysis of injury photos, injury-risk scoring across seasons, or a body-map visual — which call for a Catalyst function, an external service or a custom widget. The lesson from this project is to stay as close to no-code as the requirement allows, use Deluge for rules the builders cannot express, and treat pro-code as an extension for the few capabilities the platform lacks.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 Conclusion

This project designed and built the College Sports & Recovery Tracker on Zoho Creator to replace the informal, message-based handling of sports injuries. The application models players, injuries, recovery plans with their activities, progress entries and fitness clearances as five related forms using twelve field types, auto-numbered IDs, lookups and a subform; automates injury, overdue-recovery and high-pain alerts through three form workflows, two of them in Deluge; and presents a live dashboard with KPI counts, report links and an injury chart.

Testing in the Development environment confirmed that each functional requirement is met and that alerts fire exactly at their intended boundaries. The project also identified the next steps clearly — mandatory fields, a Physiotherapist role, a scheduled overdue check and production deployment.

9.2 Future Enhancements

- Enable mandatory fields on key inputs and configure a Physiotherapist permission profile without delete rights (Table 4.7).
- Replace the save-triggered overdue check with a daily scheduled workflow that scans all open recovery plans (BUG-02).
- Add an approval process on Fitness_Clearance so that return to play requires sign-off by the physiotherapist and the sports coordinator, or a Blueprint for the injury lifecycle.
- Add a Kanban report on Plan Status and a Calendar report on Target End Date, and include the player's name and injury details in alert e-mails.
- Publish through Stage to Production with release notes, and enable the audit trail on health-related forms.
- Integrate Zoho Cliq or WhatsApp Business for instant coach alerts, and use Zoho Analytics or a Catalyst function for season-wise injury-risk analysis.

9.3 Learning Outcomes and Career Relevance

The project turned the course content into a working application. It built practical skill in relational data modelling with lookups and subforms, in choosing field types such as

Auto Number, Image and Decision Box, in writing Deluge conditions and sendmail tasks, in configuring form workflows, and in assembling a KPI dashboard with the page builder. It also built judgement about the difference between event-driven and time-driven automation, and about when visual configuration is enough.

These skills map onto emerging roles: a **citizen developer** who automates processes in a department, a **low-code developer** who builds and maintains Zoho Creator applications, a **solutions consultant** who translates processes such as sports medicine workflows into platform configurations, and an **automation engineer** who connects such applications with other services. The experience of modelling data, automating rules and planning a governed deployment is equally valuable in conventional software engineering roles.

REFERENCES

[1] W. Rosam and O. Kalnichevski, Low-Code Development: A Practical Guide. New Delhi, India: BPB Publications.

[2] Zoho Corporation, “Zoho Creator Help Documentation.” [Online]. Available: https://www.zoho.com/creator/help/ (Accessed: 01 November 2026).

[3] Zoho Corporation, “Deluge Script Reference Guide.” [Online]. Available: https://www.zoho.com/deluge/ (Accessed: 01 November 2026).

[4] A. Sahay, A. Indamutsa, D. Di Ruscio and A. Pierantonio, “Supporting the understanding and comparison of low-code development platforms,” in Proc. 46th Euromicro Conf. Software Engineering and Advanced Applications (SEAA), 2020, pp. 171–178.

[5] A. C. Bock and U. Frank, “Low-code platform,” Business & Information Systems Engineering, vol. 63, no. 6, pp. 733–740, 2021.

[6] D. Di Ruscio, D. Kolovos, J. de Lara, A. Pierantonio, M. Tisi and M. Wimmer, “Low-code development and model-driven engineering: Two sides of the same coin?,” Software and Systems Modeling, vol. 21, no. 2, pp. 437–446, 2022.

[7] M. Hirzel, “Low-code programming models,” Communications of the ACM, vol. 66, no. 10, pp. 76–85, 2023.

[8] Gartner, Inc., “Magic Quadrant for Enterprise Low-Code Application Platforms,” Gartner Research, 2023.

[9] Zoho Corporation, “Zoho Creator Learning Centre.” [Online]. Available: https://www.zoho.com/creator/resources/ (Accessed: 01 November 2026).

APPENDIX A – ZOHO CREATOR FEATURE COVERAGE CHECKLIST

Table A.1: Feature Coverage Checklist

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
I	Application created & sections/menu organised	Y	Six navigation sections incl. dashboard	5.1	Fig. 5.1
II	Forms with ≥ 8 distinct field types	Y	5 forms, 12 field types	5.2	Fig. 5.2–5.6
II	Lookup field (relationship between forms)	Y	Player_Registration, Injury Report (×2), Recovery Plan	4.2, 5.2	Fig. 4.2, 5.3–5.6
II	Subform (parent–child data)	Y	Recovery_Activities in Recovery Plan	4.2, 5.2	Fig. 5.4
II	Field validations / layout rules	N	Typed fields only; mandatory not enabled	5.2	—
II	List report with filter, grouping, custom action	N	Default list reports only	5.3	Fig. 5.8
II	Kanban / Calendar / Timeline / Map report	N	—	5.3	—
II	Pivot table / chart report	Y	Pie – Injury Report	5.3	Fig. 5.9
II	Page / dashboard built with UI Builder	Y	Sports_Recovery_Dashboard	5.4	Fig. 5.9
II	Deluge – form events (validate / success / field actions)	Y	On success: Recovery Plan, Recovery_Progress	5.5	Code 5.1, 5.2
II	Deluge – functions, fetch, loops, collections	N	Conditions and sendmail only	5.5	—
II	Email / in-app notifications	Y	Send Mail + 2 sendmail alerts	5.5, 5.6	Fig. 5.10–5.12
II	Form workflow	Y	3 form workflows	5.6	Fig. 5.12–5.15
II	Scheduled workflow	N	—	5.6	—
II	Approval process	N	—	5.6	—
II	Blueprint	N	—	5.6	—
II	Batch / report workflow	N	—	5.6	—
II	Testing & debugging evidence	Y	Test cases, alert boundary tests	6	Table 6.1

Unit	Zoho Creator Feature	Done	Where in the Application	Report Section	Figure / Code No.
III	Integration with a Zoho service	N	Only built-in mail	5.7	—
III	Connection / REST API call (invokeurl)	N	—	5.7	—
III	Webhook or Creator REST API	N	—	5.7	—
III	Users, roles and permission profiles	N	Owner (Administrator) only	5.8	—
III	Record-level access criteria / customer portal	N	—	5.8	—
III	Zia / AI feature	N	Build Agent available in builder	5.9	Fig. 5.16
III	Analytics (charts / Zoho Analytics)	Y	KPI counts and pie chart	5.9	Fig. 5.9
IV	Performance optimisation applied & measured	N	Applied by design; not measured	7.2	Table 7.2
IV	Deployment across environments	N	Development only	7.1	Fig. 7.1
IV	Security controls (encryption, OAuth, permissions)	N	Platform authentication only	7.3	Table 7.3
IV	Governance (audit trail, administration)	N	Naming conventions and auto IDs only	7.4	—
V	Pro-code extension (Catalyst / widgets / custom functions)	N	—	5.10	—
V	Low-code limitations & spectrum analysis	Y	Report analysis	8.4, 8.5	—
—	Mobile app / portal access	N	Not demonstrated	5.11	—

Total features implemented: 12 / 32

APPENDIX B – COURSE OUTCOME MAPPING

Table B.1: Mapping of Course Outcomes to Report Chapters

CO	Course Outcome Statement	Evidenced In	Self-Assessed Level	Examiner Level
CO1	Explain the principles, architecture, and business value of low-code application development using Zoho Creator.	Ch. 1, 2	3	
CO2	Develop business applications using Zoho Creator forms, reports, workflows, and Deluge scripting.	Ch. 4, 5 (5.1–5.6), 6	2	
CO3	Integrate Zoho Creator applications with Zoho services and external systems using APIs, connections, and role-based security.	Ch. 4.6, 5.7–5.9	1	
CO4	Deploy, secure, and manage Zoho Creator applications across development and production environments following governance best practices.	Ch. 7	1	
CO5	Evaluate low-code development strategies, AI-assisted application development, and the transition from citizen developer to professional developer.	Ch. 5.9, 5.10, 8.4, 8.5, 9.3	2	

APPENDIX C – COMPLETE DELUGE SOURCE CODE

C.1 Recovery Plan – Recovery_Overdue_Alert (Created or Edited)

```
// Script ID : DS-01
// Location  : Recovery_Plan -> Created or Edited -> Successful form
submission
//             (Workflow: Recovery_Overdue_Alert)
// Purpose   : E-mail the coordinator when a plan is past its target
end date
//             and has not been marked Completed

if(Target_End_Date < zoho.currentdate && Plan_Status != "Completed")
{
    sendmail
    [
        from :"gowthamraj.m.2024.cse@rajalakshmi.edu.in"
        to :"kaileshm65552@gmail.com"
        subject :"Recovery Overdue Alert"
        message :"A recovery plan has passed its target end date and
is not yet completed."
    ]
}
```

C.2 Recovery_Progress – Recovery_Progress_Update (Created or Edited)

```
// Script ID : DS-02
// Location  : Recovery_Progress -> Created or Edited -> Successful
form submission
//             (Workflow: Recovery_Progress_Update)
// Purpose   : E-mail a high-pain alert when Pain_Level is 8 or more

if(Pain_Level >= 8)
{
    sendmail
    [
        from :"gowthamraj.m.2024.cse@rajalakshmi.edu.in"
        to :"kaileshm65552@gmail.com"
        subject :"High Pain Level Alert"
        message :"<div><span>A recovery progress entry has recorded a
pain level of 8 or above.</span></div>"
    ]
}
```

APPENDIX D – USER MANUAL

D.1 Accessing the Application

Web: open https://creatorapp.zoho.in/kaileshm65552/environment/development/college-sports-recovery-tracker in Google Chrome and sign in with the Zoho account to which the application has been shared; the navigation sections appear on the left. **Mobile:** install the Zoho Creator app from the Play Store or App Store, sign in with the same account and open "College Sports & Recovery Tracker".

D.2 Instructions for Sports Coordinator / Administrator

- Register players: open Player_Registration, fill the player's details and Submit; the Player ID is assigned automatically.
- Report an injury: open Injury Reports, select the player, fill injury date, sport, body part, type, severity, cause and description, upload a photo, set Injury Status and Submit. An alert e-mail is sent to the coordinator.
- Record fitness clearance: open Fitness_Clearance, select the injury, enter evaluation date, fitness result, recommendation, reviewer and approval status, and Submit.
- Monitor: open 1. Sports_Recovery_Dashboard to see total players, recovery plans, injuries and cleared players; use the Sports Recovery Center buttons to open any report, where records can be edited or deleted.
- Watch your inbox for "Recovery Overdue Alert" and "High Pain Level Alert" e-mails and follow up with the physiotherapist.

D.3 Instructions for Physiotherapist

- Create a recovery plan: open Recovery Plan, select the injury, enter start and target end dates, your name and plan status, then add one row per exercise in Recovery_Activities (name, minutes, sessions per week, instructions, status) and Submit.
- Log progress after each session: open Recovery_Progress, select the plan, enter the date, pain level (0–10), progress percentage, tick Exercise Completed, add

remarks and current status, and Submit. A pain level of 8 or more alerts the coordinator.

- When recovery is complete, set the plan's status to "Completed" and record the evaluation in Fitness_Clearance (available once the Physiotherapist profile is enabled — see Section 9.2).

APPENDIX E – EVALUATION RUBRIC

(For Examiner Use Only)

Student Name	GOWTHAMRAJ M	Register No.	2116240701155
Project Title	College Sports & Recovery Tracker using Zoho Creator	Date	

Table E.1: Evaluation Rubric for the Individual Project

S.No.	Criterion	Indicators	Max. Marks	Marks Awarded
1	Problem Definition & Requirement Analysis	Clarity of problem, measurable objectives, complete FR/NFR, use cases	10	
2	System Design	Architecture, ER/data model, form specifications, workflow design, permission matrix	10	
3	Forms, Reports & Pages	Variety of field types, lookups/subforms, report types, UI Builder dashboards	15	
4	Deluge Scripting & Workflows	Correctness and depth of scripts; form, schedule, approval, Blueprint, batch workflows	15	
5	Integration, RBAC & AI	Zoho/external integrations, connections, roles & permissions, Zia/analytics	10	
6	Testing, Deployment, Security & Governance	Test coverage, environments, optimisation, security controls, audit	10	
7	Live Demonstration	End-to-end working of the application across roles; feature checklist verified	15	
8	Project Report	Adherence to template and format, clarity, figures, references	5	
9	Viva-Voce	Understanding of the platform, design decisions and low-code concepts	10	
	TOTAL		**100**	

Table E.2: Performance Level Descriptors

Level	% of Max. Marks	Descriptor
Excellent	90–100	Feature implemented completely, correctly and beyond the basic requirement; student explains design choices confidently.
Good	70–89	Feature implemented correctly with minor gaps; explanation adequate.
Satisfactory	50–69	Feature partially implemented or works only for the main path; limited explanation.
Needs Improvement	< 50	Feature missing, non-functional or not understood by the student.

Signature of Internal Examiner	**Signature of External Examiner**
Signature: _____ Name: _____ Date: _____	Signature: _____ Name: _____ Date: _____

APPENDIX F – VIVA-VOCE RECORD SHEET

(For Examiner Use Only)

Student Name	GOWTHAMRAJ M	Register No.	
Department / Section		Date of Viva	
Application URL Verified	Yes / No	Demo Duration	_____ min

Table F.1: Viva-Voce Questions and Responses

Q.No.	Question / Topic Asked	Unit	Response (E / G / S / NI)	Remarks
1				
2				
3				
4				
5				
6				
7				
8				

E – Excellent G – Good S – Satisfactory NI – Needs Improvement

Table F.2: Marks Summary

Component	Max. Marks	Marks Awarded
Criteria 1–8 (Appendix E)	90	
Viva-Voce	10	
TOTAL	**100**	

Examiners' Remarks:

_____

_____

INTERNAL EXAMINER Signature: _____ Name: _____	**EXTERNAL EXAMINER** Signature: _____ Name: _____